

Formin-like 1 β phosphorylation at S1086 is necessary for secretory polarized traffic of exosomes at the immune synapse

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Abstract

T-cell receptor stimulation by antigen bound to the major histocompatibility complex (MHC) on an antigen-presenting cell (APC) induces protein kinase C (PKC) activation and the formation of the immune synapse (IS), followed by depletion of filamentous actin (F-actin) at the central region of the IS (cIS) and the polarization of multivesicular bodies (MVB) and the microtubule-organizing center (MTOC) to the IS. These events lead to polarized exosome secretion at the IS. These exosomes are involved in several crucial immune responses such as autocrine activation-induced cell death (AICD) of T lymphocytes and cytotoxicity. We analysed here how formin-like 1 β (FMNL1 β), an actin cytoskeleton-regulatory protein, regulates MTOC/MVB polarization and exosome secretion at the IS in a phosphorylation-dependent manner. IS formation was associated with transient recruitment of FMNL1 β to the IS, which was independent of protein kinase C δ (PKC δ). Simultaneous RNA interference of all FMNL1 isoforms prevented MTOC/MVB polarization and exosome secretion, which were restored by FMNL1 β WT expression. However, expression of the non-phosphorylatable mutant FMNL1 β S1086A did not restore neither MTOC/MVB polarization nor exosome secretion to control levels, supporting the crucial role of S1086 phosphorylation in MTOC/MVB polarization and exosome secretion. In contrast, the phosphomimetic mutant, FMNL1 β S1086D, restored MTOC/MVB polarization and exosome secretion. Conversely, FMNL1 β S1086D mutant did not recover the deficient MTOC/MVB polarization occurring in a PKC δ -interfered clone, indicating that S1086 FMNL1 β phosphorylation alone is not sufficient for MTOC/MVB polarization and exosome secretion. FMNL1 interference inhibited the depletion of F-actin at the cIS, which is necessary for MTOC/MVB polarization. FMNL1 β WT and FMNL1 β S1086D, but not FMNL1 β S1086A expression, restored F-actin depletion at the cIS. Thus, actin cytoskeleton reorganization at the IS underlies the effects of all these FMNL1 β variants on polarized secretory traffic. Taken together, these results point out a crucial role of S1086 phosphorylation in FMNL1 β activation, leading to cortical actin reorganization and subsequent control of MTOC/MVB polarization and exosome secretion.

eLife assessment

This **important** study uses the Jurkat T cell model to study the role of Formin-like 1 β phosphorylation at S1086 on actin dynamics and exosome release at the immunological synapse. While the evidence is **compelling** within the framework of the Jurkat model, it is limited in a broader immunological and cell-biological context due to the limitations of the model system. Jurkat is known to have a bias toward formin-mediated actin filament formation at the expense of Arp2/3-mediated branched F-actin foci observed in primary T cells. In this light, confirming major findings in primary T cells will be of importance.

Introduction

T cell receptor (TCR) stimulation by antigen bound to major histocompatibility complex (MHC) molecules on the surface of an antigen-presenting cell (APC) induces the formation of the immune synapse (IS). The IS is a specialized cell-cell interface contact area that provides a signalling platform for integration of signals leading to intercellular information exchange, in order to ensure efficient TCR signal transduction, T cell activation and the proper execution of diverse T lymphocyte effector functions (1). Comprised among these functions, IS formation triggers the convergence of T lymphocyte secretory vesicles, including multivesicular bodies (MVB) (2) (3), towards the microtubule-organizing center (MTOC) and the polarization of MTOC together with secretory vesicles to the IS (4, 5). The canonical bullseye structure of a mature IS comprises three concentric areas referred to as supramolecular activation clusters (SMACs) that can be discerned based on the specific segregation of TCR, integrins and costimulatory molecules (6). The central supramolecular activation cluster (cSMAC) is enriched in ligand-bound TCR and associated signalling molecules, the peripheral SMAC (pSMAC) in adhesion molecules such as LFA-1, and the outer distal SMAC (dSMAC) in filamentous actin (F-actin) and CD45 (7) (8). IS assembly and subsequent secretory responses are coordinated by both actin and microtubule cytoskeletons, whose interplay regulates the polarized secretory response towards the IS (9) (7). IS formation is associated with an initial increase in cortical actin at the IS (10), followed by a decrease in cortical actin density at the central region of the immune synapse (cIS) including the cSMAC, that contains the secretory domain (11) (12). The partial depletion of F-actin at the cIS has been proposed, apart from allowing the focusing and docking of secretory vesicles for secretion (10), to initiate key subsequent events leading to MTOC polarization and delivery of secretory vesicles, such as lytic granules in cytotoxic T lymphocytes (CTL) (12) and cytokine-containing secretory vesicles in T-helper (Th) lymphocytes (13), to the secretory domain of the IS. The CTL degranulation of diverse secretory lytic granules with MVB structure results in the secretion to the synaptic cleft of Fas ligand (FasL)-containing exosomes (14) (15) (16) (17) (2) (18). These exosomes, along with perforin and granzymes, which are secreted in both soluble and nanoparticulate form in so-called supramolecular attack particles (SMAPs) (19) (20), lead to the induction of target cell apoptosis. In addition, FasL-containing exosomes have been involved in autocrine FasL/Fas dependent, activation-induced cell death (AICD) produced upon TCR triggering (16) (21) (17) (22), an important immunoregulatory event involved in the downregulation of T cell immune responses (23) (24).

We have previously described that cortical actin reorganization at the IS plays an important role in MVB polarized traffic leading to exosome secretion in Th lymphocytes (25) (26) (3). With respect to the molecular cues controlling this process, we have shown that TCR-stimulated protein kinase C δ (PKC δ) regulates cortical actin reorganization at the IS, thereby controlling MTOC/MVB polarization and ultimately leading to exosome secretion at the IS and AICD in Th lymphocytes

(25 [↗](#)). Moreover, PKC δ is necessary for the polarization of lytic granules and the induction of cytotoxicity by mouse CTL (27 [↗](#), 28 [↗](#)), which supports a general role of PKC δ in secretory traffic leading to apoptosis in T lymphocytes. Besides PKC δ , several actin cytoskeleton regulators, including the formin FMNL1 and Diaphanous-1 (Dia1), also regulate MTOC polarization (29 [↗](#)) (30 [↗](#)) (31 [↗](#)). In this context, we have shown both PKC δ -dependent phosphorylation of FMNL1 at the IS and PKC δ -dependent F-actin clearing at the cIS, which participate in MTOC/MVB polarization leading to exosome secretion in CD4⁺ Jurkat T lymphocytes forming IS (25 [↗](#)) (26 [↗](#)) (3 [↗](#)). However, a formal connection between FMNL1 phosphorylation, F-actin regulation and MVB polarization/exosome secretion at the IS has not been established yet.

Regarding potential formin regulatory pathways, we have shown that FMNL1, but not Dial, is strongly phosphorylated in T lymphocytes by PKC δ activators such as phorbol myristate acetate (PMA), but also upon TCR stimulation via an anti-TCR agonist as well as IS formation (26 [↗](#)). FMNL1 phosphorylation was inhibited in PKC δ -interfered T lymphocytes (26 [↗](#)), which is associated with a deficient MTOC polarization towards the IS (25 [↗](#)), supporting a PKC δ role in both FMNL1 phosphorylation and MTOC polarization (25 [↗](#)) (26 [↗](#)). Interestingly FMNL2, a related formin which exhibits a high homology with FMNL1, is phosphorylated by PKC α and, to a lower extent, by PKC δ , at S1072 (32 [↗](#)). This phosphorylation reverses FMNL2 autoinhibition mediated by interaction of N-terminal Diaphanous inhibitory domain (DID) with the C-terminal Diaphanous autoinhibitory domain (DAD), resulting in increased F-actin assembly, β 1-integrin endocytosis, invasive motility (32 [↗](#)) and filopodia elongation (33 [↗](#)). In this regard, recent evidence suggests that the mutation of a particular residue in FMNL2 impacting the DID-DAD interaction may result in functional implications for the characteristic actin-regulating activity of FMNL2, specifically in the context of podosome formation in macrophages (34 [↗](#)). Out of the three FMNL1 isoforms (α , β and γ) present in T lymphocytes and Jurkat cells (35 [↗](#)), S1086 in FMNL1 β is surrounded by a sequence displaying high homology to the one surrounding S1072 in FMNL2 (32 [↗](#)) (26 [↗](#)) (Fig. 1 [↗](#)). Moreover, we have shown that FMNL1 β is the only FMNL1 isoform phosphorylated upon IS formation and capable of recovering MTOC polarization when it was reexpressed in cells simultaneously interfered for the three FMNL1 isoforms (26 [↗](#)). Thus, we hypothesize that IS-induced, PKC δ -dependent phosphorylation at S1086, a residue located within FMNL1 β DAD, may release the autoinhibition mediated by DID/DAD-interaction, thereby leading to the activation of FMNL1 β . Consequently, this activation would mediate the F-actin reorganization at the IS and the MTOC/MVB polarization, thereby enabling the subsequent exosome secretion. Here, we report that FMNL1 β phosphorylation at S1086 regulates F-actin reorganization leading to MTOC/MVB polarization and exosome secretion.

Results

FMNL1 interference and YFP-FMNL1 β variants expression

According to our previous data (26 [↗](#)), FMNL1 β plays a crucial role in MTOC polarization towards the IS. Moreover, we have also shown that FMNL1 β is strongly phosphorylated in T lymphocytes by PKC δ activators such as PMA, as well as by TCR stimulation and upon IS formation, which could potentially be related to FMNL1 β activation (26 [↗](#)). As previously stated, FMNL2 becomes active upon PKC α phosphorylation at S1072, enhancing F-actin assembly (32 [↗](#)) (33 [↗](#)). In FMNL1 β , Si086 in arginine-rich DAD (labelled in Fig. 1 [↗](#)) is surrounded by a sequence displaying high similarity to that around S1072 in FMNL2 (36 [↗](#)) (32 [↗](#)) (26 [↗](#)) (Fig. 1A [↗](#)). Thus, we hypothesized that phosphorylation of FMNL1 β at Si086 may release DID-DAD auto-inhibition, thereby activating FMNL1 β and regulating MTOC polarization. To address this point, we introduced Si086A and Si086D mutations in the shFMNL1-HA-YFP-FMNL1 β construct (35 [↗](#)) (26 [↗](#)), which act as non-phosphorylatable and phosphomimetic residues, respectively (Fig. 1B [↗](#)). These resulting bicistronic vectors produce both FMNL1 interference and the corresponding YFP-FMNL1 β variant expression (35 [↗](#)), acting as useful tools to analyse the role of FMNL1 β in IS formation during

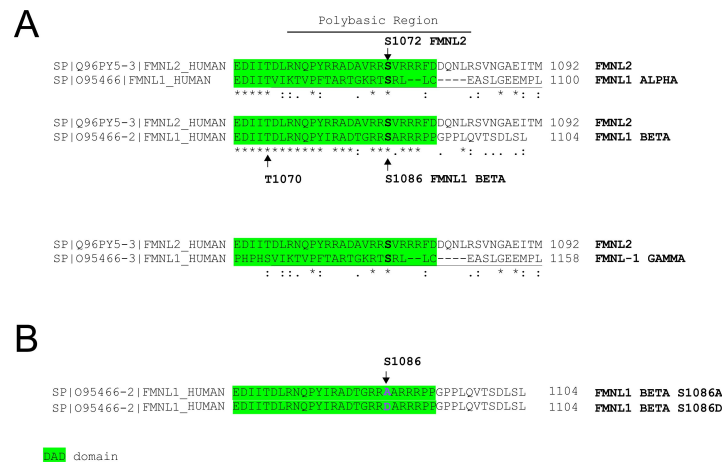


Fig. 1

C-terminal alignment of FMNL1 isoforms. S1086 of FMNL1 β shows a high degree of homology with PKC-phosphorylated S1072 in FMNL2.

Panel A) amino acid sequences of the C termini of FMNL2, FMNL1 α and FMNL1 β , as well as FMNL1 γ containing a C-terminal intron retention, and sharing the final C-terminal amino acids with FMNL1 α (shown underlined). The three FMNL1 isoforms share identical sequence (100% identity) from amino acid residue 1 to T1070, and diverge in the C-terminal region, which includes the DAD autoinhibitory domain (36 [\[36\]](#)). The DAD domain sequence, which is responsible for autoinhibition in the murine homolog is highlighted in green (36 [\[36\]](#)), and the identical C-terminal amino acids shared by FMNL1 α and FMNL1 γ are underlined. The arginine-rich, polybasic region common to other formins such as FMNL2 (32 [\[32\]](#)), FHOD3 (56 [\[56\]](#)) and FHOD1 (55 [\[55\]](#)) (see Discussion) is also shown overlined. Panel B), C-terminal amino acid sequences of the two different point mutations at S1086 in FMNL1 β used in this study.

transient expression experiments (26). To confirm the previously described efficacy of these vectors (33) (26), we first analysed endogenous FMNL1 interference and YFP-FMNL1 β variants expression in C3 Jurkat clone transfected with the different vectors at the single cell level by immunofluorescence, using an anti-FMNL1 antibody that recognizes all its isoforms. This analysis was performed in transfected Jurkat cells forming synapses, to rule out any potential effect of IS formation and subsequent T lymphocyte activation on both endogenous FMNL1 interference and YFP-FMNL1 β variants expression. As seen in **Fig. 2A**, C3 cells transfected with the FMNL1 interfering plasmid shFMNL1-YFP (YFP⁺ cells) (second row), showed very low anti-FMNL1 signal when compared with non-transfected cells in the same preparation or control YFP⁻ cells (first row). In contrast, cells transfected with YFP-FMNL1 β WT, YFP-FMNL1 β S1086A or YFP-FMNL1 β S1086D (rows three to five, respectively), showed higher anti-FMNL1 fluorescence signals when compared to nontransfected cells in the same microscopy field (**Fig. 2A**). The **Suppl. Fig. S1B** shows, by single cell image analysis, that the mean fluorescence intensity (MFI) of the anti-FMNL1 in shFMNL1-YFP-expressing cells is extremely low when compared to the control YFP⁻ group, whereas it is much higher in YFP-FMNL1 β WT, YFP-FMNL1 β S1086A or YFP-FMNL1 β S1086D-expressing cells. As shown in **Suppl. Fig. S1A**, anti-FMNL1 MFI correlated with YFP-FMNL1p MFI in YFP-FMNL1 β WT, YFP-FMNL1pS1086A or YFP-FMNL1 β S1086D-expressing cells. To further verify this data, lysates from bulk populations of transfected cells were analysed by western blot (WB) using the same anti-FMNL1 antibody. The apparent molecular weights (MW) of the bands observed in the WB analysis were compatible with the predicted MW of endogenous FMNL1 (150 kDa) and the chimeric YFP-FMNL1 β variants (150 + 30 kDa) (**Fig. 2B**). The endogenous FMNL1 (150 kDa band) levels in lysates from cells transfected with YFP-FMNL1 β WT, YFP-FMNL1pS1086A or YFP-FMNL1 β S1086D were lower than in control untransfected cells (**Fig. 2B**). The reduction of endogenous FMNL1 expression, as assessed by WB (between 40–80% reduction relative to control YFP⁻ cells), was apparently lower than that observed by single cell imaging analysis (compare panels A and B in **Fig. 2** with **Suppl. Fig. S1B**). This is not surprising, since the transfection efficiency was relatively low (20–50%) and thus cell populations analysed by WB contained transfected as well as untransfected cells.

YFP-FMNL1 β WT is phosphorylated upon PKC activation but neither YFP-FMNL1 β S1086A nor YFP-S1086D are

Once the different YFP-FMNL1 β variants were expressed, we analysed the ability of a PKC δ activator to induce their phosphorylation. To this end, we stimulated C3 clone transfected with the different constructs with the PKC δ activator PMA, which has been shown to induce strong FMNL1 phosphorylation (26). Subsequently, we immunoprecipitated FMNL1 with an anti-FMNL1 that recognises all FMNL1 isoforms (26) and analysed these immunoprecipitates (IPs) by WB with anti-phospho-Ser PKC substrate (32) (**Fig. 3**). In all the IPs from PMA-stimulated cells, anti-phospho-Ser PKC substrate detected a band corresponding to endogenous FMNL1 (150 kDa), which was 8- to 30-fold more intense than in IPs from unstimulated cells (**Fig. 3**, lower graph). In IPs from PMA-stimulated cells expressing YFP-FMNL1 β WT, anti-phospho-Ser PKC substrate detected an additional band of the predicted MW (150 + 30 kDa), which was also recognized by anti-FMNL1 (**Fig. 3**). However, in IPs from PMA-stimulated cells expressing YFP-FMNL1 β S1086A or S1086D no apparent phosphorylation of these FMNL1 β chimeras could be detected (**Fig. 3**). Thus, substitution of S1086 by a non-phosphorylatable or a phosphomimetic residue fully abolished PKC δ activator-induced phosphorylation of FMNL1 β , as detected by anti-phospho-Ser PKC substrate, confirming our hypothesis that S1086 is indeed the residue that is phosphorylated in FMNL1 β upon PKC activation.

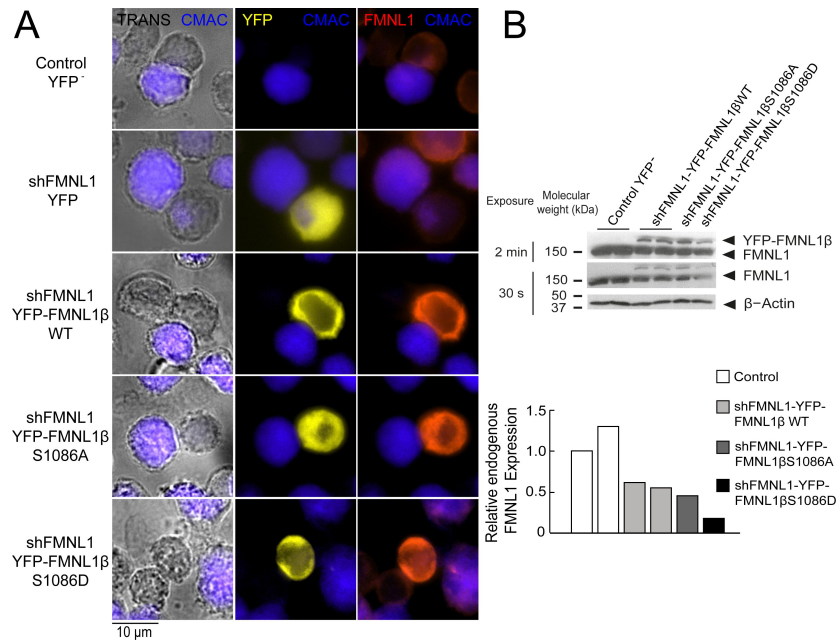


Fig. 2.

Expression of YFP-FMNL1β S1086 variants in FMNL1-silenced cells.

Panel A), C3 control clone was untransfected (Control YFP⁻) (first row) or transfected with FMNL1-interfering (shFMNL1-HA-YFP) (second row), or FMNL1-interfering expressing interference-resistant YFP-FMNL1βWT (shFMNL1-HA-YFP-FMNL1βWT) (third row), YFP-FMNL1βS1086A (fourth row) or YFP-FMNL1βS1086D (fifth row) constructs. Subsequently, cells were challenged with CMAC-labelled SEE-pulsed Raji cells (blue) for 1 h, fixed, stained with anti-FMNL1, and imaged by epifluorescence microscopy. Representative MIP images of merged transmittance (TRANS), CMAC (blue), YFP (yellow), and anti-FMNL1 (red) channels are indicated for the different cell groups. Panel B), cell lysates corresponding to the indicated cell groups were analysed by WB developed with anti-FMNL1 antibody (two different expositions). The lower bar graph depicts the WB quantification showing the endogenous FMNL1 expression in the different cell groups relative to control untransfected (YFP⁻) cells. Results are representative of data from several experiments (n=3) with similar results.

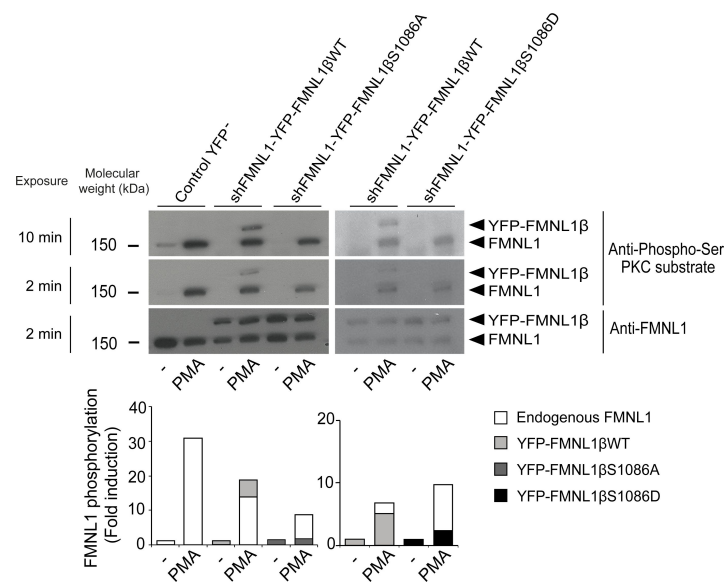


Fig. 3

S1086 in FMNL1 β is phosphorylated upon PKC activation.

C3 control clone was untransfected (Control YFP⁻) or transfected with either FMNL1- interfering expressing interference-resistant YFP-FMNL1 β WT (shFMNL1-HA-YFP-FMNL1 β WT), YFP-FMNL1 β S1086A or YFP-FMNL1 β S1086D constructs. Subsequently, cells were stimulated or not (-) with PKC δ activator PMA for 30 min. The different cell groups were lysed and immunoprecipitated with anti-FMNL1. These IPs were analysed by WB, first with anti-Phospho-Ser PKC substrate antibody (two different expositions) and then reprobed with anti-FMNL1 to normalize Phospho-Ser PKC substrate signal for FMNL1 protein levels. The lower graph represents the normalized fold induction of phosphorylation of the different FMNL1 variants. Results are representative of data from several experiments (n=3) with similar results.

S1086 phosphorylation of FMNL1 β is crucial for MTOC/MVB polarization towards the IS

Next, we wanted to assess whether S1086 phosphorylation was related to FMNL1 β regulation of MTOC/MVB polarization towards the IS. To this end, untransfected (control YFP⁻) C3 cells, or transfected with the plasmids interfering all FMNL1 isoforms and expressing YFP (shFMNL1-YFP), YFP-FMNL1 β WT, YFP-FMNL1 β S1086A, or YFP-FMNL1 β S1086D (35) were challenged with SEE-pulsed Raji cells (blue). Next, we quantified MTOC/MVB polarization index (PI) in fixed synapses formed by the mentioned cell groups by epifluorescence microscopy and deconvolution, as previously described (37) (25) (26) (Suppl. Fig. S2A, Fig. 4). In the first series of experiments, average MTOC PI of untransfected (Control YFP⁻) C3 cells forming synapses with unpulsed Raji cells was significantly lower than with SEE-pulsed Raji cells, and similar to MTOC PI of FMNL1-interfered cells (shFMNL1-YFP) forming synapses with SEE-pulsed Raji cells (Suppl. Fig. S2B). This confirms that antigenic stimulation at the IS indeed triggers MTOC polarization and validates our synapse model measurements. In addition, confirming our previous data (26), MTOC/MVB polarization was disrupted upon FMNL1 interference to the levels obtained with unpulsed Raji cells (Fig. 4A, compare the first two rows, Fig. 4B, Suppl. Fig. S2B), and was restored by YFP-FMNL1 β WT expression (Fig. 4A, third row and 4B). Interestingly, non-phosphorylatable YFP-FMNL1 β S1086A expression was unable to restore MVB/MTOC polarization to control levels (Fig. 4A, fourth row and 4B), whereas phosphomimetic YFP-FMNL1 β S1086D expression restored MVB/MTOC polarization to the levels achieved by YFP-FMNL1 β WT expression (Fig. 4A, fifth row and 4B). Comparable results were obtained by confocal microscopy (Suppl. Fig. S3). From confocal images, *en face* views of the IS interface were generated (Suppl. Fig. S4, S5 and Suppl. Videos 1 and 2). These images show the polarized MTOC or the accumulation of MVB, respectively, in the cIS region of untransfected (control YFP⁻) C3 cells or cells expressing YFP-FMNL1WT or YFP-FMNL1S1086D, but not in cells transfected with shFMNL1-YFP or expressing YFP-FMNL1 β S1086A. Thus, phosphorylation of S1086 in FMNL1 β appears to be essential for secretory polarized traffic towards the IS in T lymphocytes. Moreover, throughout the experiment, we observed that MTOC and MVB were polarizing together towards the IS. This fact, together with the previous work regarding segregation between MTOC movement and secretory granules traffic (13) (28, 38, 39), prompted us to simultaneously analyse both MTOC and MVB positioning with respect to the IS at the single cell level. Remarkably, we observed a robust linear correlations between MVB and MTOC PIs both in untransfected (control YFP⁻) C3 cells and cells transfected with shFMNL1-YFP (Pearson's linear correlation coefficients 0.96 and 0.95, respectively, Suppl. Fig. S6). Furthermore, it is noteworthy that MVB and MTOC centers of mass (MVB^C and MTOC^C, respectively) are very closely located in all the studied cell groups, regardless of polarization state (Fig. 4, Suppl. Fig. S3). Although the MTOC and MVB did not efficiently polarize in cells transfected with shFMNL1-YFP, their MVB^C were still located next to MTOC^C (Fig. 4A, second row), and equivalent results were obtained in all the analysed cell groups (Fig. 4). Thus, phosphorylation of S1086 in FMNL1 β appears to be essential for both MTOC and MVB polarization towards the IS in T lymphocytes.

YFP-FMNL1 β S1086D phosphomimetic mutant expression does not rescue MTOC polarization in PKC δ -interfered T lymphocytes

The former results supported the contribution of FMNL1 β and its phosphorylation at S1086 to MVB/MTOC polarization, but did not address the sufficiency in this process of PKC δ -controlled, S1086 FMNL1 β phosphorylation. In this context, we have previously shown that FMNL1 β lies downstream of PKC δ in the same pathway controlling MTOC/MVB polarization (26) (3). Thus, in order to analyse this important issue we transfected the PKC δ -interfered P5 clone with the different plasmids and compared the MTOC PI among the different cell groups, as previously done in C3 clone. We have previously shown that neither PKC δ nor FMNL1 interference affects IS

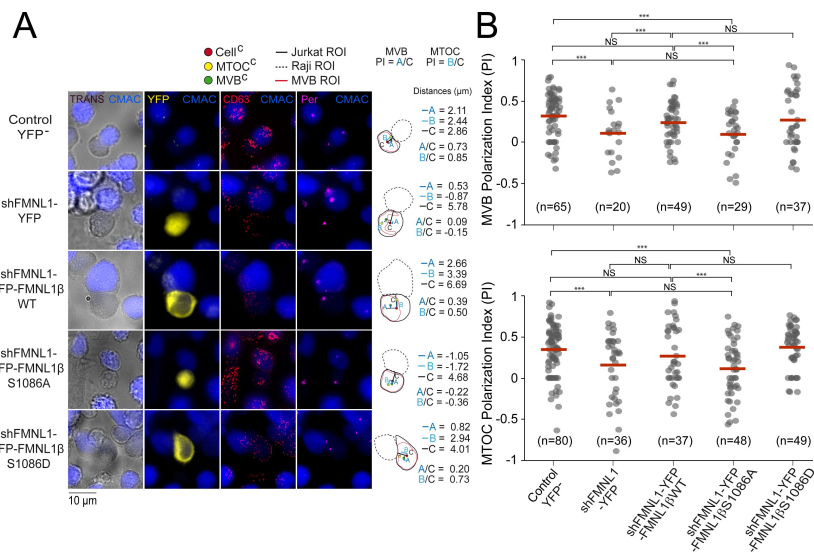


Fig. 4

FMNL1β phosphorylation at S1086 is involved in MTOC/MVB polarization towards the immune synapse.

C3 control clone was untransfected (Control YFP⁻) (first row) or transfected with FMNL1-interfering (shFMNL1-HA-YFP) (second row), or FMNL1-interfering expressing interference-resistant YFP-FMNL1βWT (shFMNL1-HA-YFP-FMNL1βWT) (third row), YFP-FMNL1βS1086A (fourth row) or YFP-FMNL1βS1086D (fifth row) constructs. Subsequently, cells were challenged with CMAC-labelled, SEE-pulsed Raji cells (blue) for 1 h, fixed, stained with anti-pericentrin (magenta) to label the MTOC and anti-CD63 (red) to label MVB, and imaged by epifluorescence microscopy. Panel A), representative MIP images with the indicated merged channels for each of the specified cell groups, along with a schematic diagram on the right representing the measured parameters used to calculate the MTOC and MVB PI. This includes the distance in microns between the MTOC (or MVB^C) projection on the vector defined by the Cell^C-synapse axis and the Cell^C ("B" or "A" distance, respectively), and the distance between the Cell^C and the synapse ("C" distance). Panel B), dot plots of MVB and MTOC PI from each of the indicated cell groups, corresponding to the indicated number of synapses from a experiment similar to that described in panel A) are depicted. NS, not significant. ***, p ≤ 0.05. Results are representative of data from several experiments (n=3) with similar results.

conjugate formation, since P5 clone and FMNL1-interfered C3 clone formed IS with SEE-pulsed Raji cells (25) (26). In addition, IS formed by P5 clone with SEE-pulsed Raji cells (fifth group in **Suppl. Fig. S2B**) had a MTOC PI comparable to those of IS formed by C3 clone with unpulsed Raji cells or P5 clone with unpulsed Raji cells (first and fourth group, respectively, in **Suppl. Fig S2B**), supporting that both FMNL1 or PKC δ interference fully inhibited MTOC polarization.

In contrast to what was found in the C3 clone (**Fig. 4** and **Suppl. Fig. S3**), YFP-FMNL1 β WT expression in PKC δ and FMNL1-interfered P5 cells did not restore MTOC PI to the values observed in control YFP⁻ C3 cells (**Fig. 5**), which is compatible with the idea that FMNL1 β lies downstream of PKC δ and its phosphorylation is indispensable for MTOC polarization. Consistently, expression of YFP-FMNL1 β S1086A, non-phosphorylatable by PKC, was also unable to enhance MTOC PI to the values observed in control YFP⁻ C3 cells (**Fig. 5B**). Moreover, phosphomimetic YFP-FMNL1 β S1086D expression, which restored MVB/MTOC PI to the values achieved by YFP-FMNL1 β WT expression in the C3 clone (**Fig. 4** and **Suppl. Fig. S3**), did not restore MTOC PI in PKC δ and FMNL1-interfered P5 cells to the levels observed in control YFP⁻ C3 cells (**Fig. 5B**). Taken together, these results show that although FMNL1 β phosphorylation at S1086 is necessary, it does not seem to be sufficient for MTOC polarization, at least in cells lacking PKC δ .

FMNL1 β translocation to the immune synapse is independent of S1086 phosphorylation and PKC δ

Previous results have shown that FMNL1, apart from being mainly located at the cytosol and centrosomal areas, was also found located at the IS in a small percentage of synaptic conjugates, in end-point IS experiments (29) (26). If confirmed, the possibility of an IS-induced FMNL1 translocation to the IS may provide the molecular basis underlying a potential regulatory effect of FMNL1 on cortical actin cytoskeleton reorganization at the IS as well as on MTOC/MVB and secretion granule polarization, occurring both in CTL and Th cells, as suggested by several authors (9, 12) (13) (40) (3). However, although it could be inferred (29) (41) that the active translocation of any FMNL1 isoform from cytosol towards the IS is induced upon IS formation, it has not been formally demonstrated yet (29) (41). This is probably due to the fact that most early studies have used an end-point approach that does not allow to analyse the incipient IS (29) (42) (43) (44). Since FMNL1 β is responsible for MTOC/MVB polarization to the IS (44), we analysed FMNL1 β subcellular location in developing synapses by time-lapse, epifluorescence microscopy (43) (44). For this analysis, FMNL1-interfered, YFP-FMNL1 β WT-expressing C3 clone was challenged with SEE-pulsed Raji cells. As evidenced in the example shown in **Suppl. Video 3** (upper panel) and **Fig. 6A** (upper row), a transient relocation of initially cytosolic YFP-FMNL1 β WT to the IS (white arrow) was observed in these cells. The average ($\pm$ SD) duration of YFP-FMNL1 β WT accumulation at the IS in C3 clone was around 6 min (6 min 18 s $\pm$ 1 min 34 s, n=6 synapses).

Considering the major role of PKC α in FMNL2 relocation from the plasma membrane to the endosomal membrane (32), the high similarity between FMNL2 and FMNL1 β C-terminal regions (**Fig. 1**) and the reported PKC δ -mediated FMNL1 β phosphorylation (26), we next aimed to analyse the role of PKC δ in YFP-FMNL1 β WT translocation to the IS. To this end we analysed YFP-FMNL1 β WT translocation upon IS formation in the PKC δ -interfered P5 clone (25) (**Suppl. Video 3** and **Fig. 6A**). The results in P5 clone showed a clear translocation of YFP-FMNL1 β WT to the IS (white arrow), with an accumulation of around 8 minutes at the IS (8 min 55 s $\pm$ 3 min 2 s, n=6 synapses, not significant differences with the C3 clone), so YFP-FMNL1 β WT translocation to the IS in P5 and C3 clones showed comparable kinetics.

Moreover, to directly analyse the specific involvement of S1086 phosphorylation in this process, we challenged FMNL1-interfered, YFP-FMNL1 β S1086A and YFP-FMNL1 β S1086D-expressing C3 cells with SEE-pulsed Raji cells. As shown in **Suppl. Video 4** and **Suppl. Fig. S7**, both YFP-

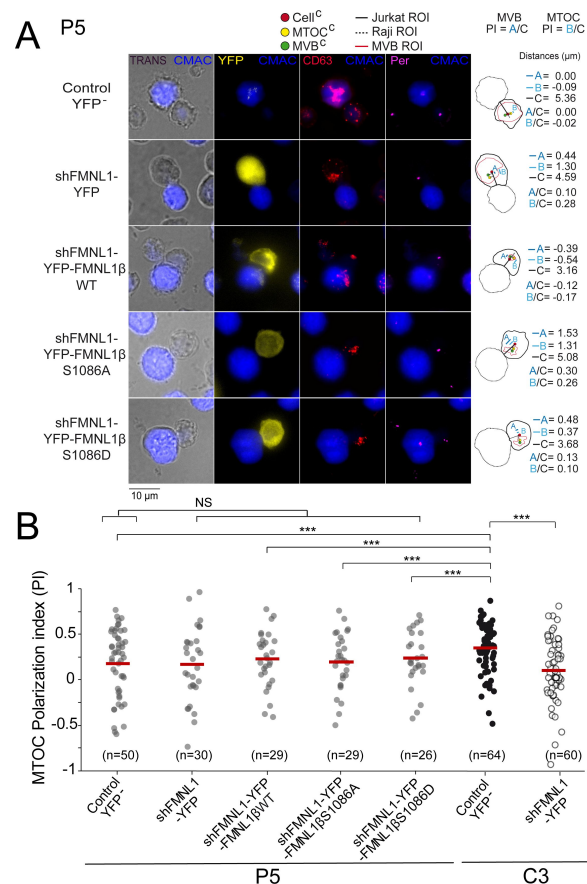


Fig. 5.

YFP-FMNL1βS1086D expression does not rescue deficient MTOC polarization in PKCδ-interfered cells.

PKCδ-interfered P5 clone was untransfected (Control YFP⁻) (first row) or transfected with FMNL1-interfering (shFMNL1-HA-YFP) (second row), FMNL1-interfering expressing interference-resistant YFP-FMNL1βWT (shFMNL1-HA-YFP-FMNL1βWT) (third row), YFP-FMNL1βS1086A (fourth row) or YFP-FMNL1βS1086D (fifth row) constructs. In parallel, C3 control clone was untransfected (Control YFP⁻) or transfected with FMNL1-interfering (shFMNL1-HA-YFP) construct. Subsequently, cells were challenged with CMAC-labelled, SEEpulsed Raji cells (blue) for 1 h, fixed, stained with anti-pericentrin (magenta) to label the MTOC and anti-CD63 (red) to label MVB, and imaged by epifluorescence microscopy. Panel A), representative MIP images with the indicated merged channels for each of the specified cell groups along with their corresponding diagrams as in **Fig. 4**, representing the measured parameters used for calculating the MTOC and MVB PI are shown. This includes the distance in microns between the MTOC^C(or MVB^C) projection on the vector defined by the Cell^C-synapse axis and the Cell^C(“B” or “A” distance, respectively), and the distance between the Cell^C and the synapse (“C” distance). Panel B), dot plots of MTOC PI of each cell group. Untransfected (Control YFP⁻) and FMNL1-interfering (shFMNL1-HA-YFP) C3 clone synapses corresponding to the data shown in **Suppl. Fig. S3B** were included in the far-right columns, as a reference. NS, not significant; ***, $p \leq 0.05$. Results are representative of data from several experiments ($n=3$) with similar results.

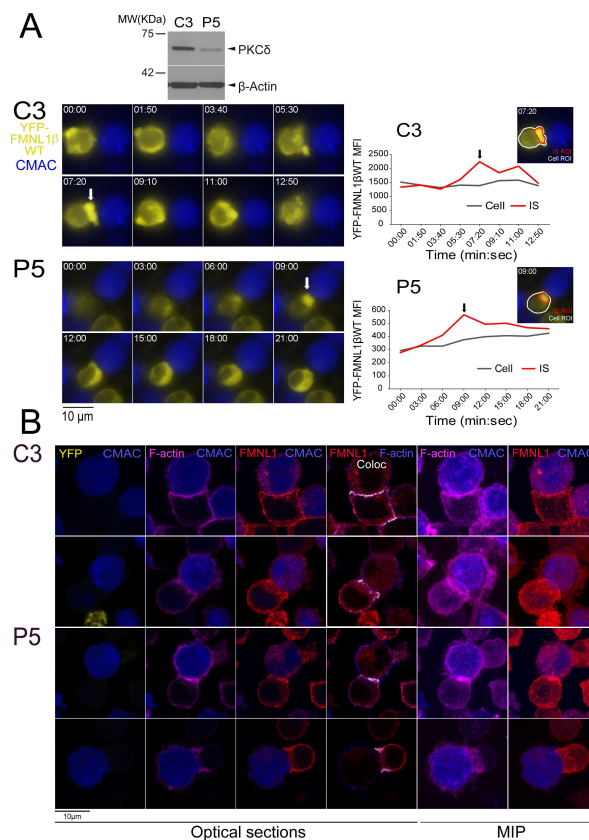


Fig. 6.

FMNL1 β recruitment to the immune synapse is PKC δ -independent.

C3 control and P5 PKC δ -interfered cells were transfected with FMNL1-interfering expressing interference-resistant YFP-FMNL1 β WT (shFMNL1-HA-YFP-FMNL1 β WT) plasmid. Subsequently, both transfected clones were simultaneously challenged with CMAC-labelled, SEE-pulsed Raji cells (blue) attached to slides and time-lapse acquisition of emerging synapses was performed as indicated in Materials and Methods. The videos (7 fps) ([Suppl. Video 3](#)) were captured and in panel A), left, representative frames from videos of each clone are shown. White arrows indicate accumulations of YFP-FMNL1 β WT at the IS. WB analysis of cell lysates from both clones (top inset) shows PKC δ silencing in P5 clone. In the right side, YFP-FMNL1 β WT MFI within the cell ROI (grey line) and the IS ROI (red line) are represented. The inserts in the graphs include the cell ROI (white) and the IS ROI (red) used for the time-lapse measurements on representative frames for both clones. Results are representative of the data from several videos ($n=6$ for each clone) with similar results. Panel B), control C3 (upper rows) and P5 PKC δ -interfered cells (lower rows) were simultaneously challenged with CMAC-labelled SEE-pulsed Raji cells (blue) attached to slides. After 1 h, synapses were fixed and immunofluorescence developed with anti-FMNL1 (red) to label endogenous FMNL1 and phalloidin (magenta) to label F-actin. Synapses were imaged with confocal fluorescence microscopy and colocalization pixels between FMNL1 (red) and F-actin (acquired in magenta, changed to blue in the fourth column) are represented in white. Representative optical sections of synapses formed by both clones are shown in the left columns. The colocalization coefficients were: First row, Pearson= 0.50, Manders= 0.877; Second row, Pearson = 0.431, Manders= 0.864; Third row: Pearson= 0.434, Manders= 0.838; Fourth row, Pearson= 0.484, Manders= 0.794. MIP images of the same synapses are shown in the two far right columns. Results are representative of data from several experiments ($n=3$) with similar results.

FMNL1 β S1086A and YFP-FMNL1 β S1086D transiently relocated to the IS. The average time of YFP-FMNL1 β S1086A accumulation at the IS was around 7 min (7 min 30 s $\pm$ 1 min 58 s, n=6 synapses), whereas YFP-FMNL1 β S1086D showed an average accumulation time of around 6 min (6 min 52 s $\pm$ 2 min 9 s, n=7 synapses; not significant differences between YFP-FMNL1 β S1086A and YFP-FMNL1 β S1086D, nor between any of the mutants and YFP-FMNL1 β WT). Therefore, these results confirm that the translocation and accumulation of FMNL1 β in the IS region of Jurkat cells is a process that occurs independently of both PKC δ and S1086 FMNL1 β phosphorylation.

A more detailed study of subsynaptic location of endogenous FMNL1 in C3 control clone using fixed synapses demonstrated that FMNL1 accumulated at the cIS (**Fig. 6B**, first row) but also at the edges of the synapse, around the dSMAC, both in C3 control clone (**Fig. 6B**, first and second rows) and in P5 PKC δ -interfered clone (**Fig. 6B**, third and fourth rows). At these subsynaptic locations, FMNL1 colocalized with F-actin (white pixels), which is compatible with the idea that FMNL1 regulates F-actin at the IS.

Subsequently, we studied the subsynaptic locations of YFP-FMNL1 β WT, YFP-FMNL1 β S1086A, and YFP-FMNL1 β S1086D in C3 control clone, to identify whether S1086 phosphorylation affected these locations. The results from fixed synapses, obtained through both epifluorescence (**Suppl. Fig. S7** and **S8**) and confocal microscopy (**Suppl. Fig. S9**), revealed the presence of accumulations of the three YFP-FMNL1 β variants in the IS. Additionally, in **Suppl. Fig. S9**, it can be observed that the three YFP-FMNL1 β variants colocalized with F-actin at the IS.

Taken together, these observations show that, upon IS formation, FMNL1 β translocates to the IS, with both PKC δ and S1086 phosphorylation being dispensable for the translocation to the IS of both endogenous FMNL1 and YFP-FMNL1 β variants.

S1086 phosphorylation of FMNL1 β is necessary for cortical F-actin reorganization during MTOC/MVB polarization

The observations that YFP-FMNL1 β translocated to the IS, that cortical F-actin remodelling was considered to be sufficient for centrosome polarization in CTL (12), and that cortical F-actin remodelling was associated with MTOC and secretory granule polarization in CD4⁺ cells (13) (45), prompted us to study a potential role of FMNL1 β and its phosphorylation at S1086 in synaptic F-actin architecture. To address this point, fixed synapses formed by the C3 clone transfected with the previously described plasmids were stained with phalloidin AF647 to label F-actin and imaged by confocal fluorescence microscopy. The relative area of the F-actin-low region at the cIS (Fact-low cIS area / IS area) in the IS interface, which mirrors cortical F-actin reorganization at the IS, was measured as indicated in Materials and Methods and represented for all cell groups. As seen in a representative example in **Fig. 7A**, the dot plot in **Fig. 7B** and **Suppl. Video 5**, the F-actin-low region at the cIS was smaller in FMNL1-interfered cells than in untransfected control YFP⁻ cells (**Fig. 7A**, second and first columns, respectively), as it was also in PKC δ -interfered cells (25) (26). Of note, YFP-FMNL1 β WT expression restored F-actin depletion at the cIS to control levels (**Fig. 7B**). In contrast, YFP-FMNL1 β S1086A expression did not rescue F-actin depletion to control levels (**Fig. 7B**), whereas YFP-FMNL1 β S1086D did rescue F-actin depletion to control levels (**Fig. 7B**). Thus, FMNL1 β phosphorylation at S1086 appears to be necessary for F-actin reorganization at the IS, as it was for MTOC/MVB polarization (**Fig. 4**).

Three-dimensional FMNL1 β distribution at the synapse

We observed a positive effect of FMNL1 β phosphorylation at S1086 on cortical F-actin rearrangement at the IS (former paragraph and **Fig. 7**) and YFP-FMNL1 β translocated to the central IS, but also to the dSMAC (**Suppl. Video 3** and **4**, **Fig. 6** and **Suppl. Fig. S7**, **S8** and **S9**). To investigate this further, we analysed in more detail the subsynaptic distribution of the endogenous FMNL1, YFP-FMNL1 β WT, and the YFP-FMNL1 β S1086A and S1086D mutants with

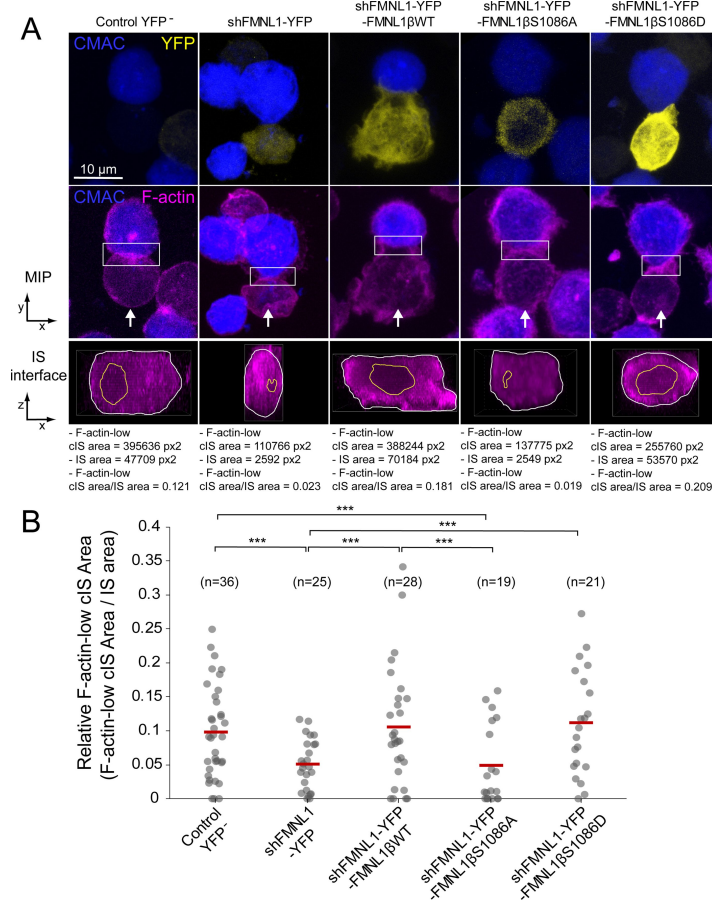


Fig. 7.

FMNL1β phosphorylation at S1086 regulates the F-actin architecture of the immune synapse.

C3 control clone was untransfected (Control YFP⁻) (first column) or transfected with FMNL1-interfering (shFMNL1-HA-YFP) (second column), FMNL1- interfering expressing interference-resistant YFP-FMNL1βWT (shFMNL1-HA-YFP-FMNL1βWT) (third column), YFP-FMNL1βS1086A (fourth column) or YFP-FMNL1βS1086D (fifth column) constructs. Subsequently, cells were challenged with CMAC-labelled, SEE-pulsed Raji cells (blue) for 1 h, fixed, stained with phalloidin AF647 (magenta) to label F-actin, and imaged by confocal microscopy. Panel A), the upper rows display the top, yx views corresponding to the MIP images of the specified, merged channels of a representative example from each of the indicated cell groups. White arrows indicate the direction to visualize the *en face* views of the IS (IS interface) enclosed by the ROIs (white rectangles), as shown in **Suppl. Video 5**. In the lower panels, the enlarged ROIs (2x zoom) used to generate the IS interface, zx images of each cell group are shown. The areas of the F-actin-low region at the cIS (Fact-low cIS area) (yellow line) and the synapse (IS area) (white line) were defined and measured as indicated in Materials and Methods, and the relative area of the F-actin-low region at the cIS (Fact-low cIS area / IS area) was calculated and represented. Panel B), relative area (Fact-low cIS area / IS area) dot plot distributions and average area ratios (red horizontal lines) for the indicated number of IS conjugates developed by each cell group are shown. This figure is related to **Suppl. Video 5**. NS, not significant; ***, $p \leq 0.05$. Results are representative of data from several experiments (n=3) with similar results.

respect to the actin cytoskeleton. To this end, fixed synapses formed by the C3 clone transfected with the different plasmids were stained with phalloidin, anti-FMNL1 and anti-phospho-Ser PKC substrate antibody and imaged by confocal fluorescence microscopy (**Suppl. Videos 6** and **7**, **Fig. 8** and **Suppl. Fig. S10**). **Figs. 8A and 8B** show representative examples of synaptic conjugates yx views (first row) and cropped synapse areas (white rectangles in first row) used to generate zx IS interfaces (second row), as shown in **Suppl. Video 6** and **7**. Subsequently, the colocalization pixels (white) along all the Z-stack at the IS interface (interface colocalization) were determined as indicated in Materials and Methods. This involved merging the indicated fluorescence channels (**Fig. 8A**, second row, F-actin in blue and anti-FMNL1 in red; **Fig. 8B**, second row, F-actin in blue and YFP fluorescence in green) on the IS interfaces of the synaptic areas generated as shown in **Suppl. Video 6**. MFI profile of each channel (F-actin in blue, anti-FMNL1 in red, YFP in green and colocalization in grey) along the indicated white arrows are shown below each IS interfaces. As shown in **Fig. 8**, and consistently with results from **Fig. 7**, both control YFP⁻ (first column) and YFP-FMNL1 β WT-expressing cells (third and fourth columns) exhibited a wide F-actin depletion area at the cIS and an F-actin accumulation at the dSMAC. Interestingly, in some synapses from control YFP⁻ (first column) and YFP-FMNL1 β WT-expressing cells (fourth column), F-actin and FMNL1 colocalized (white pixels) at certain small areas at the wide F-actin-low region at the cIS, although the MFI profiles show relatively less F-actin and FMNL1 at the cIS than at the dSMAC. In other synapses from both control YFP (not shown) and YFP-FMNL1 β WT-expressing cells (third column), F-actin and FMNL1 colocalization occurred at the F-actin rich-area corresponding to the dSMAC (**Fig. 8A**), which was consistent with the data from **Fig. 6B**. In contrast, and consistently with **Fig. 7**, FMNL1-interfered cells (**Fig. 8**, second column) displayed small F-actin low areas at the cIS. Moreover, synapses from YFP-FMNL1 β S1086A-expressing cells (fifth column) did not exhibit the F-actin depletion at the cIS, whereas synapses from the YFP-FMNL1 β S1086D-expressing cells (sixth column) exhibited wide F-actin depletion at the cIS, comparable to those observed in the control YFP- and in the YFP-FMNL1 β WT-expressing cells (**Fig. 8**). In cells expressing the YFP-FMNL1 β variants, colocalization of all the YFP-FMNL1 β variants with F-actin was mainly in the F-actin low area at the cIS or at the dSMAC. When colocalization in all Z optical sections was assessed using directly the YFP fluorescence construction instead of anti-FMNL1 signal (**Fig. 8B**, second row, F-actin in blue and YFP fluorescence in green), comparable results were obtained. In addition, we analysed the subsynaptic location of FMNL1 with respect to Ser-phosphorylated PKC substrates by using an anti-FMNL1 and an anti-phospho-Ser PKC substrate. Both endogenous FMNL1 in control YFP-cells (**Suppl. Video 7** and **Suppl. Fig. S10A**, first column) and YFP-FMNL1 β WT in YFP-FMNL1 β WT+ cells, (**Suppl. Fig. S10A**, third column) strongly and specifically colocalized with anti-phospho-Ser PKC substrate at the cIS. This is evidenced by the accumulation of white, colocalization pixels at low FMNL1 density areas located at the cIS, which is also an F-actin-low area (**Fig. 8**, see MFI profiles for control YFP⁻ and YFP-FMNL1 β WT+ cells). When colocalization at the synapse interface was assessed using directly the YFP fluorescence construction instead of anti-FMNL1 signal (**Suppl. Fig. S10B** second row, anti-phospho-Ser PKC substrate in blue and YFP construction in green and **Suppl. Video 7**) comparable results were obtained. In addition, as expected and as a negative control, no colocalization of FMNL1 and anti-phospho-Ser PKC substrate was observed in FMNL1-interfered cells (**Suppl. Fig. S10** and **Suppl. Video 7**, second column), which is compatible with the idea that both endogenous FMNL1 and YFP-FMNL1 β WT are specifically phosphorylated at the cIS.

S1086 phosphorylation of FMNL1 β regulates exosome secretion

The aforementioned results support that FMNL1 β and its phosphorylation at S1086 are involved in MTOC/MVB polarized traffic to the IS, most probably via control of F-actin rearrangement at the IS. MVB transport and fusion to the plasma membrane are necessary for exosome secretion (46) (22) and PKC δ controls F-actin depletion at the cIS (25), thus, it is conceivable that FMNL1 β may control subsequent exosome secretion at the IS. To directly analyse FMNL1 β contribution to exosome secretion, we used an exosome secretion reporter assay. We transiently cotransfected C3

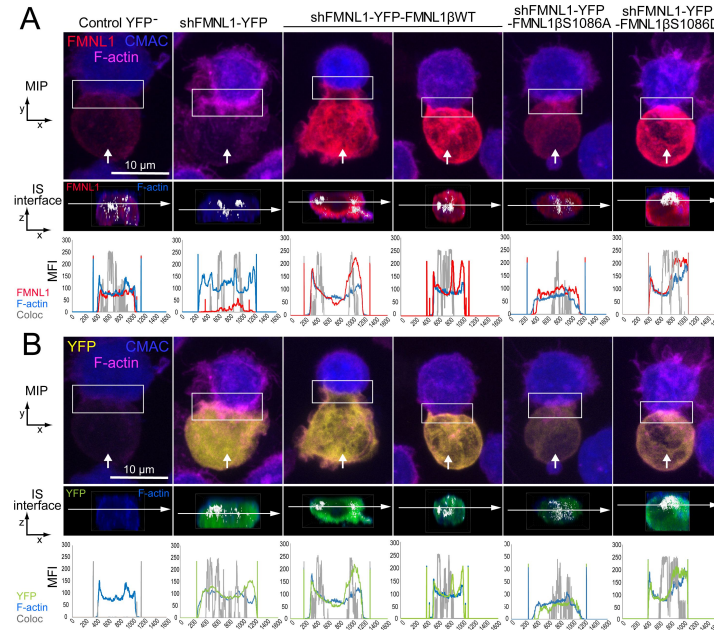


Fig. 8.

Three-dimensional distribution and colocalization of FMNL1 and F-actin at the immune synapse interface.

C3 control clone was untransfected (Control YFP⁻) (first column) or transfected with FMNL1-interfering (shFMNL1-HA-YFP) (second column), FMNL1-interfering expressing interference-resistant YFP-FMNL1βWT (shFMNL1- HA-YFP-FMNL1βWT) (third and fourth column), YFP-FMNL1βS1086A (fifth column) or YFP-FMNL1βS1086D (sixth column) constructs. Subsequently, cells were challenged with CMAC-labelled SEE-pulsed Raji cells (blue) for 1 h, fixed, and stained with phalloidin (magenta) and anti-FMNL1 (red). The corresponding shFMNL1 construction is in yellow, and synaptic conjugates were imaged by confocal fluorescence microscopy. Please realize that for the IS interface and since interface colocalization in NIS-AR only works with red, green and blue channels, F-actin (acquired in magenta) was changed to blue color, and YFP (acquired in yellow) was changed to green in the second row of each panel. Panel A), the upper row includes top, yx views corresponding to the MIP images of representative examples of each cell group. Vertical white arrows indicate the direction to visualize the *en face* views of the IS (IS interface) enclosed by the ROIs (white rectangles), as shown in [Suppl. Video 6](#). In the second row, the enlarged ROIs (2x zoom) used to generate the IS interface, zx images of each cell group are shown. Subsequently, interface colocalization pixels (white) were generated by merging the indicated channels in the second row of each panel (F-actin in blue merged to anti-FMNL1 in red), at the IS interfaces of the synaptic areas (generated as shown in [Suppl. Video 6](#)). The last frame of these videos corresponds to the *en face* view (interface) (second row in both panels). MFI profiles along the indicated line (horizontal white arrow) of each separate channel (F-actin in blue, anti-FMNL1 in red) and the colocalization pixels (grey) are shown below the IS interfaces. Panel B), same as panel A, but the top views show YFP-expressing constructs signal instead of the anti-FMNL1 signal. The IS interfaces and the MFI profiles show F-actin (magenta changed to blue) and YFP (yellow changed to green). This figure is related to [Suppl. Video 6](#). At least 6 synapses of each cell group were analysed. Results are representative of data from several experiments (n=3) with similar results.

clone with the exosome reporter GFP-CD63 expression plasmid (2 [↗](#)) (47 [↗](#)) and the different shFMNL1-YFP constructs. Subsequently, IS were formed between the cotransfected C3 clone and SEE-pulsed Raji cells, exosomes were purified from cell culture supernatants (2 [↗](#)), lysed, and exosome lysates were analysed by anti-CD63 WB. This secretion assay excludes the detection of exosomes released by Raji cells in the coculture, that otherwise may mask the exosomes released by Jurkat cells (2 [↗](#)) (22 [↗](#)). In parallel, we determined nanovesicle concentrations in coculture supernatants by using Nanoparticle Tracking Analyses (NTA) (Fig. 9A [↗](#)). Anti-CD63 WB analysis of exosome secretion is shown in Fig. 9B [↗](#). We normalized exosomal GFP-CD63 signal to the transfection efficiency of the exosome reporter (measured by GFP-CD63 signal from cell lysates) and the number of viable exosome-secreting cells (Fig. 9B [↗](#)). As shown in Fig. 9 [↗](#), interference with all FMNL1 isoforms not only reduced exosomal GFP-CD63 levels, but also reduced SEE-stimulated nanovesicle concentration increase. These results mirrored those obtained in cells in PKC δ -interfered cells (25 [↗](#)). Moreover, interference with all FMNL1 isoforms and YFP-FMNL1 β WT expression recovered exosome secretion to the levels produced by control YFP⁻ cells. However, YFP-FMNL1 β S1086A expression did not recovered exosome secretion, whereas YFP-FMNL1 β S1086D expression rescued exosome secretion to the values observed in control YFP⁻ or YFP-FMNL1 β WT-expressing cells (Fig. 9B [↗](#)). Taken together, these results on exosome secretion endorse the outcomes regarding FMNL1 interference and FMNL1 β - mediated control of cortical F-actin reorganization at the IS and their role on MVB polarized secretory traffic.

Discussion

Our previously published results suggest that PKC δ -dependent phosphorylation of FMNL1 β may regulate its function in F-actin reorganization at the IS and hence affect MTOC polarization (25 [↗](#)) (26 [↗](#)). Here, we have identified for the first time a positive regulatory role of FMNL1 β phosphorylation at S1086 in cortical F-actin regulation that governs MTOC/MVB polarization and exosome secretion at the IS in Th lymphocytes. Moreover, the fact that the phosphomimetic YFP-FMNL1 β S1086D expression in PKC δ and FMNL1-interfered cells, did not recover the MTOC PI observed in C3 control cells (Fig. 5 [↗](#)) supports the idea that FMNL1 β phosphorylation at S1086 is necessary but not sufficient for MTOC polarization, at least in cells lacking PKC δ . Thus, these results support that another signal, apart from PKC δ activation, or another PKC δ -controlled pathway not involving FMNL1 β phosphorylation, evoked upon TCR triggering at the IS, is also implicated in MTOC/MVB polarization to the IS.

Regarding actin cytoskeleton regulation and the molecular mechanisms involved in MTOC polarization, it is remarkable that, while cortical F-actin remodelling has been considered to be sufficient for centrosome polarization, at least in CTL (12 [↗](#)), several lines of evidence indicate that, in certain conditions (such as absence of Arp2/3), MTOC polarization may occur normally in the absence of any cortical F-actin reorganization (29 [↗](#)). In addition, several studies indicated that the MTOC is an F-actin organizing center (48 [↗](#)), and F-actin depletion around the MTOC crucially facilitates MTOC polarization to the IS in B-cell receptor (BCR)-stimulated B lymphocytes (49 [↗](#)) (50 [↗](#)). More recently, these studies have been extended to MTOC polarization in T lymphocytes, since F-actin clearance from the centrosomal area has been recently shown to occur during IS formation with APC (26 [↗](#)) (51 [↗](#)). Taking all these observations together, it is clear that MTOC and MVB polarization induced by IS formation are regulated by HS1/WASp/Arp2/3-dependent cortical and non-cortical centrosomal actin networks, and that PKC δ appears to coordinately regulate the reorganization of both actin networks via paxillin phosphorylation at T538 and FMNL1 β phosphorylation at S1086 (3 [↗](#)) (26 [↗](#)) (51 [↗](#)) (52 [↗](#)). Thus, TCR-controlled, PKC δ -induced paxillin phosphorylation at T538 could be the abovementioned additional PKC δ -controlled pathway not involving FMNL1 β phosphorylation that may collaborate with phosphorylated FMNL1 β in MTOC polarization. In this context, the existence of an F-actin network around the centrosome (48 [↗](#)) involved in MTOC polarization to the IS (49 [↗](#)) (26 [↗](#)) has added a new level of complexity to the regulation of this process (52 [↗](#)). The observation of the impact of

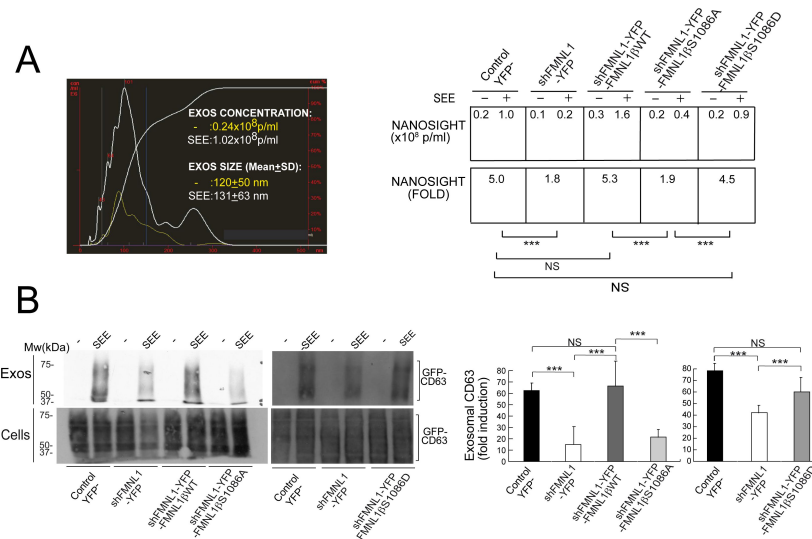


Fig. 9.

FMNL1 β phosphorylation at S1086 regulates exosome secretion at the immune synapse.

C3 control clone was transfected with exosome reporter GFP-CD63 alone (Control YFP⁻) or co-transfected with the exosome reporter GFP-CD63 and 3-fold molar excess of either FMNL1-interfering (shFMNL1-HA-YFP), or FMNL1-interfering expressing interference-resistant YFP-FMNL1 β WT (shFMNL1-HA-YFP-FMNL1 β WT), YFP-FMNL1 β S1086A or YFP-FMNL1 β S1086D constructs. Subsequently, the different cell groups were challenged with unpulsed (-) or SEE-pulsed Raji cells for 4 h and exosomes were collected and purified from cell culture supernatants and analysed by NTA and WB, as indicated in Materials and Methods. Panel A), left, NTA analyses corresponding to nanovesicles isolated from cell culture supernatants of control YFP⁻ Jurkat cells stimulated with unpulsed Raji cells (-) or SEE-pulsed Raji cells (SEE). Right, concentration and SEE-stimulated nanoparticle concentration increase (fold) induction of secreted nanovesicles from several NTA analyses for the indicated cell groups. Panel B), left, WB analyses of exosomal GFP-CD63 reporter isolated from the indicated cell culture supernatants. Right, quantification of normalized, fold induction of exosomal GFP-CD63 secretion from three experiments similar to the one described in the left panel. NS, not significant; ***, $p \leq 0.05$. Results are representative of data from several experiments ($n=3$) with similar results.

PKC δ - mediated phosphorylation of S1086 in YFP-FMNL1 β WT on the recovery of the F-actin low area at the cIS ([Fig. 7](#)), associated with the restoration of MTOC polarization ([Fig. 4](#)), does not exclude the possibility that S1086 phosphorylation in FMNL1 β may also play a role in the clearance of F-actin around the MTOC, collaborating with the depletion of cortical F-actin at the cIS in MTOC polarization. Thus, more experiments are needed to address the possibility that S1086 phosphorylation in FMNL1 β may affect centrosomal area F-actin, together with cortical F-actin, since reorganization of these two distinct, F-actin networks collaborate in MTOC polarization ([26](#)) ([49](#)) ([52](#)) ([53](#)).

In addition, a further level of complexity regarding FMNL1 activities and their regulation in the context of T cell activation relies on the existence of three different FMNL1 isoforms in T lymphocytes (α , β and γ) ([35](#)), showing full sequence identity from N-terminus up to T1070, but diverging C-terminal domains ([Fig. 1](#)) ([36](#)). These differences most probably underpin the diverse subcellular distribution of the three isoforms, which appear to distribute differentially within the cytosol, MTOC, cortical actin at the IS and Golgi, exerting distinct roles in the functions and subcellular distribution of these subcellular organelles ([29](#)) ([35](#)) ([26](#)) (this article). In this context, it is important to clarify that, out of the three FMNL1 isoforms, FMNL1 β is the only one capable of restoring MTOC polarization when reexpressed, and the only one that undergoes PKC δ -mediated phosphorylation upon PKC activation and IS formation ([26](#)). This illustrates the finely tuned and specific involvement of FMNL1 β in secretory polarized traffic in T lymphocytes.

Our results on FMNL1 β phosphorylation are certainly related to what was found in formin homology domain protein 1 (FHOD1) regulation. FHOD1 is grouped into Dia-related formins ([54](#)). FHOD1 requires active Rac to be recruited to the plasma membrane, but its recruitment appears insufficient for FHOD1 activation, which is achieved only after being phosphorylated by Rho-dependent protein kinase (ROCK) at three serine and threonine sites included in a polybasic arginine-rich region inside the C-terminal DAD ([55](#)), leading to disruption of FHOD1 autoinhibitory state and F-actin stress fiber formation ([55](#)). This activation process is similar to what we have described in this article for FMNL1 β . Moreover, the PKC-dependent mode of FMNL2 activation during integrin activation ([32](#)) and filopodia formation ([33](#)) is comparable to what we have found in FMNL1 β during polarized traffic to the IS. Furthermore, FHOD3 formin is also autoinhibited by an intramolecular interaction between its C- and N-terminal domains. Phosphorylation of the three highly conserved residues (S406, S412, and T416) within the polybasic, C-terminus of FHOD3 by ROCK1/2 is sufficient for its activation and is crucial in regulating myofibrillogenesis in cardiomyocytes ([56](#)). Thus, with the results provided here for FMNL1 p, it has been reported that at least four different formins are activated through phosphorylation by three different kinases at the C-terminal polybasic regions, which are included in a common DAD. This underscores the evolutionary importance of this regulatory mechanism in formin activation, ultimately leading to very diverse cellular responses. However, although we do not have conclusive evidence supporting that PKC directly phosphorylates FMNL1 β , the presence of a conserved S1086 within a RRSV/AR motif that contains a potential target for classical PKCs ([32](#), [33](#), [57](#)), along with the recognition of the phosphorylated residue by a validated anti-Phospho-Ser PKC substrate antibody ([32](#)), certainly support this possibility.

With regards to potential regulatory pathways involved in formin activation and subcellular localization, it is remarkable that the most abundant formins in T lymphocytes, Dia1 and FMNL1 ([29](#)), are constitutively inactive in the cytoplasm due to intramolecular DAD-DID binding, which blocks formin ability to nucleate and elongate actin filaments ([7](#)). Regarding a possible control of formin subcellular location by PKC, it has been shown that both relocation of FMNL2 from the plasma membrane to the endosomal compartment and activation are dependent on C-terminal S1072 phosphorylation by PKC α ([32](#)). However, FMNL2 location in filopodia seems to be independent of the phosphorylation state ([33](#)). We have previously described PKC δ - dependent phosphorylation of FMNL1 β ([26](#)), and here we have here demonstrated that this phosphorylation occurs at S1086, leading to F-actin reorganization at the IS, and resulting in MTOC

polarization. However, in contrast to FMNL2 relocation, which is dependent on PKC α , we have shown that transient YFP-FMNL1 β WT relocation from cytosol to the IS occurred in the absence of PKC δ (Fig. 6), in conditions where neither F-actin reorganization (25) nor MTOC polarization (Fig. 5) occurred. Moreover, both YFP-FMNL1 β S1086A and YFP-FMNL1 β S1086D translocated to the IS (Suppl. Video 4 and Suppl. Fig. S7), demonstrating that S1086 phosphorylation is not required for IS location. However, as it happens during S1072 phosphorylation in FMNL2 (32), it is possible that blocking phosphorylation at S1086 through the introduction of single point mutations (YFP-FMNL1 β S1086A or YFP-FMNL1 β S1086D) may not block FMNL1p translocation to the IS per se, but may alter the kinetics of the process. For this reason, in future approaches, a thorough study on the kinetic considerations on S1086 PKC δ -mediated phosphorylation contribution to FMNL1p translocation to the IS will need to be addressed. Regarding FMNL1p translocation, the early reports (29) did not demonstrate IS-induced FMNL1p relocation, most probably because the establishment of a mature, fully productive IS, is intrinsically a stochastic, rapid and asynchronous process difficult to image at its initial stages (43). Moreover, most early reports used an end-point approach that forces cell conjugate formation by centrifugation (42), and this approach does not allow to analyse incipient IS (42) (43) (44). We have circumvented this caveat by using living cell imaging of spontaneously formed, emerging IS at high temporal resolution (43). In this manner, we have observed that the average duration of YFP-FMNL1 β WT accumulation at the IS is short, around 6 min. The low fraction of synaptic conjugates displaying endogenous FMNL1 at the synapse in endpoint experiments (approximately 10%, not shown) is consistent with such a short duration for transient YFP-FMNL1 β WT translocation to the IS. We have performed some preliminary experiments analyzing the subcellular localization of FMNL1 in chimeric antigen receptor-bearing T lymphocytes (CAR T) with dual targeting of CD19/CD22, forming synapses with CD19⁺/CD22⁺ Raji cells used as CTL targets. We observed, in fixed-cell images, the presence of endogenous FMNL1 in the IS region of CAR T cells, which also exhibited MTOC polarization towards the synapse (not shown). This observation extends our results on FMNL1 localization at the IS in a synapse model, supporting that it may also occur in synapses developed by primary human T lymphocytes.

Although the mechanism(s) involved in FMNL1 β recruitment to the IS remains unclear, it is probably that it involves Rac1 and Cdc42 small G proteins (30) (29) (58), which are activated at the IS upon TCR triggering (59). However, we cannot exclude the participation of another PKC isoform or a different protein kinase that may induce FMNL1 β phosphorylation in different residues upon previous FMNL1 β relocation to the IS primed by small G protein, as was shown for FMNL2 relocation (32). Further experiments are needed to establish this important point.

Taken together, our results support that IS-induced, PKC δ -dependent phosphorylation of S1086 in FMNL1 β DAD, most probably executed by PKC δ itself, activates FMNL1 β , which in turn regulates cortical actin at the IS and hence controls MTOC/MVB polarization and exosome secretion. The behaviour of non-phosphorylatable YFP-FMNL1 β S1086A (equivalent to FMNL2-S1072A described in (32)) confirms this hypothesis, since its expression inhibits F-actin reorganization at the IS, MTOC polarization and exosome secretion. In this respect, it has been reported that the phosphorylation of FMNL2 by PKC α in S1072 is necessary for the secretion of the pro-metastatic factor Angiopoietin-like 4 (ANGPTL4) in breast cancer cells (60), which constitutes the first description for an activated formin involvement in secretion during cell invasion. In addition, ANGPTL4 can be secreted into exosomes by lung cancer cells (61) (62) and FMNL2 controls exosome secretion (63). Thus, although it is not clear whether ANGPTL4 is released in exosomes or in soluble form (60), both FMNL2 and FMNL1 β appear to control PKC-regulated exosome secretion. However, the extracellular signals evoking exosome secretion are strikingly different in the various cell types, since in tumor cells hypoxia and radiation can amplify the otherwise constitutive and multidirectional exosome secretion, whereas T lymphocytes constitutively secrete very low levels of exosomes and none in a polarized manner unless activated through TCR or synaptic contact (22). Taken together, these considerations support a more general role of the

formin-actin cytoskeleton axis in exosome secretion. Given all these considerations, it will be interesting to analyse the role of FMNL2 in inducible exosome secretion at the IS in T lymphocytes, but also the role of FMNL1 β in exosome secretion by cancer cells during cell invasion. This is particularly relevant as exosome secretion is required for directionally persistent and efficient *in vivo* movement of cancer cells (64), and it is known that FMNL1 stimulates both leukemia cell proliferation and migration (58) (65).

The fact that FMNL1- and PKC δ -interfered cells have a similar phenotype to that of FMNL1- or PKC δ -interfered cells (26) (25) (this paper), supports the notion that PKC δ and FMNL1 β participate in the same F-actin regulatory pathway controlling MTOC/MVB polarization. Additional experiments involving *in vitro* measurements of F-actin reorganizing/severing activities of the diverse FMNL1 β variants, phosphorylated or not *in vitro* by PKC δ , will be necessary to address whether PKC δ directly phosphorylates FMNL1 β at S1086.

Our findings show that FMNL1 colocalizes with F-actin at the IS edges which are part of the dSMAC (Figs. 6, 8). This data are aligned with the previous observation that Dia1 and FMNL1 are intrinsically inactive because they undergo intramolecular folding and remain in the cytoplasm (7) and that after IS formation, they are enriched at the tips of F-actomyosin spikes at the outer edge of the dSMAC, giving rise to the actomyosin arcs at the pSMAC in the IS formed by Jurkat cells (41) (7). Dia1 is not a target of PKC, whereas FMNL1 is phosphorylated by PKC activation (26). Thus, in this paper we have focused on FMNL1. High resolution, spatio-temporal analysis of FMNL1 β localization at the different subsynaptic locations, together with superresolution techniques (41) (43) may provide new clues regarding the subsynaptic F-actin network targeted by phosphorylated FMNL1 β . Even although the nature of the downstream effectors and consequences of FMNL1 β phosphorylation on synaptic F-actin architecture can only be speculated at this point, concentric actomyosin arcs generated by FMNL1 propel TCR microclusters towards the IS and reinforce cytotoxicity (41) (7). Thus, high-resolution spatiotemporal analysis of actomyosin arcs structure and function during TCR microcluster movement in FMNL1-interfered cells, and/or cells expressing the FMNL1 β mutants described here, will provide important clues on this important matter. It is noteworthy that a majority of the image analysis of IS organization and F-actin dynamics is based on planar synapses obtained using artificial APC substitutes, by either coating stimulatory and co-stimulatory molecules on glass or plastic surfaces or by embedding these molecules into lipid bilayers (66) (7) (53). These artificial synapses can indeed be imaged and analysed with the highest possible resolution (67) (41) (43). However, the cell-cell synapse model analysed here mimics the complex interactions and irregular, 3D stimulatory surface of a physiological synapse better than artificial synapses do (66) (7). In this important sense, the data presented here are indeed closer to a physiological situation. In addition, since actomyosin arcs arise from active, formin-dependent nucleation sites at the outer edge of the dSMAC (41), it will be interesting to analyse, using superresolution techniques, whether phosphorylated FMNL1 β is located at these sites. Indeed, actomyosin arc network spanning from dSMAC along the pSMAC is the most likely source of T cellbased force that augments cytotoxicity by straining the target cell plasma membrane (7). Thus, it will be interesting to study the effect of FMNL1 β mutants on actomyosin arcs formation and analyse functional deficiencies (i.e., decreased T cell-APC adhesion frequency, IS size and IS formation efficiency) in IS formed by cells interfered in FMNL1 β and/or expressing the FMNL1 β variants described here. Visualization of actin and actomyosin networks, TCR microclusters and secretion granules dynamics in “real” live T cell-APC conjugates at the superresolution level (7) is clearly needed to advance in this area.

Remarkably, although it has already been shown that FMNL1 is important for MTOC polarization in Jurkat cells and for target cell lysis in CTL (29), our data provide the first molecular basis underpinning this essential FMNL1 function. Here, we demonstrate that FMNL1 β and its phosphorylation at S1086 participate in F-actin-controlled, polarized MVB traffic secreting effector exosomes (2, 17), which deliver proapoptotic signals to target cells (19) (68) (20). In addition, FMNL1 promotes proliferation and migration of leukemia cells (58) and mediates

posterior perinuclear actin polymerization to promote T lymphocyte effector cell migration to inflammatory sites to enable T cell-mediated autoimmunity (65 [↗](#)). PKC δ is crucial for directional T cell migration (69 [↗](#)) (70 [↗](#)) and this process requires MVB polarized traffic and exosome secretion in the direction of migration (64 [↗](#)). For that reason, it will be interesting to analyse the effects of the FMNL1 p mutants we have described here during T lymphocyte migration and tumorigenesis, as well as to study the role of FMNL1 β phosphorylation in several autoimmune disorders.

In addition, improving our understanding of the molecular bases underlying the traffic events involved in polarized secretion of pro-apoptotic exosomes as we have performed here will provide clues to modify crucial immune functions involving apoptosis, such as cytotoxicity by CTLs and immunoregulatory AICD and their associated pathologies (17 [↗](#)) (71 [↗](#)) (2 [↗](#)) (18 [↗](#)) (25 [↗](#)) (72 [↗](#)) (73 [↗](#)).

Materials and methods

Cells

Raji B cell line was obtained from the ATCC. Cell lines were cultured in RPMI 1640 medium containing L-glutamine (Invitrogen) with 10% heat-inactivated FCS (Gibco) and penicillin/streptomycin (Gibco). Jurkat control (C3 clone) and PKC δ -interfered (P5 clone) stable clones have already been described and were cultured with puromycin (25 [↗](#)).

Plasmids and transient transfection

pEGFP-C1CD63 (expressing GFP-CD63) was provided by G. Griffiths. CD63 is an enriched marker in intraluminal vesicles (ILVs) contained into MVB and exosomes and has been widely used to study MVB secretory traffic in both living and fixed cells (2 [↗](#), 17 [↗](#)). The FMNL1 interfering vector (shFMNL1-HA-YFP) and FMNL1-interfering, YFP-FMNL1 β WT expressing vector (shFMNL1-HA-YFP-FMNL1 β WT) were previously described (35 [↗](#)) and generously provided by Dr. Billadeau. shFMNL1-HA-YFP-FMNL1 β S1086A and shFMNL1-HA-YFP-FMNL1 β S1086D mutants were generated by site-directed mutagenesis, as previously reported (74 [↗](#)). Briefly, overlap extension PCR was used to introduce the desired mutations into a 900 bp SalI-NotI fragment, which was then used to replace the corresponding wild type sequence in shFMNL1-HA-YFP-FMNL1 β —WT. All amplifications were carried out with Platinum SuperFi DNA polymerase (ThermoFisher). Primers used were: SalI-FMNL1 β _F (CCAGAGCCTGGATGCGCTGTGG), FMNL1 p-S 1086A_F (CAGACACAGGCCGCGCTGCCGTCGGCGTCCC), FMNL1p-S1086A_R (GGGACGCCGACGGCAGCGCGGCGCCTGTGTCTGC), FMNL1p-S1086D_F (CAGACACAGGCCGCGGATGCCGTCGGCGTCCC), FMNL1p-S1086D_R (GGGACGCCGACGGGCATCGCGGCGGCGCCTGTGTCTGC), and NotI-FMNL1p_R (GCGAGCTCTAGGGCCGCTTGCG). Presence of the desired mutations, and absence of undesired ones, was confirmed by Sanger DNA sequencing (Eurofins Genomics, LightRun). Both interference and expression of the different chimeric molecules were assessed in single cells level by immunofluorescence and/or bulk cell populations by WB with an anti-FMNL1 antibody, which recognizes all FMNL1 isoforms (see below). Jurkat clones were transiently transfected with 20–30 μ g of the plasmids as described (17 [↗](#)). To analyse the effects of shFMNL1 constructs on exosome secretion by Jurkat cells expressing exosome reporter GFP-CD63, Jurkat cells were transiently co-transfected with a three-fold molar excess of the different shFMNL1-YFP plasmids with respect to the GFP-CD63 exosome-reporter plasmid, to favor that the exosomes analysed were mainly those secreted by Jurkat cells containing both the GFP-CD63 plasmid and the shFMNL1 construct (2 [↗](#)) (18 [↗](#)).

Antibodies and reagents

Rabbit monoclonal anti-human PKC δ EP1486Y for WB that does not recognize mouse PKC δ (Abcam). Rabbit monoclonal anti-PKC δ EPR17075 for WB that recognizes both human and mouse PKC δ (Abcam). Mouse monoclonal anti-human CD3 (clone UCHT1) for cell stimulation and immunofluorescence (BD Biosciences and Santa Cruz Biotechnology). Mouse monoclonal anti-FMNL1 clone C-5 for WB and immunofluorescence and mouse monoclonal anti-FMNL1 clone A-4 for immunoprecipitation (Santa Cruz Biotechnology), recognizing all FMNL1 isoforms. Mouse monoclonal anti- γ -tubulin for immunofluorescence (SIGMA, clone GTU-88).

Rabbit polyclonal anti-pericentrin ab4448 for immunofluorescence (Abcam). Mouse monoclonal anti-human CD63 (clone TEA3/18) for immunofluorescence (Immunostep). Mouse monoclonal anti-CD63 clone NKI-C-3 for WB (Oncogene). Rabbit polyclonal Phospho-Ser PKC substrate antibody for WB and immunofluorescence (Cell Signalling Technology, #2261). Fluorochrome-coupled secondary antibodies (goat-anti-mouse IgG AF488 A-11029, goat-anti-rabbit IgG AF488 A-11034, goat-anti-mouse IgG AF546 A-11030, goat-anti-mouse IgG AF647 A-21236) for immunofluorescence (ThermoFisher). Horseradish peroxidase (HRP)-coupled secondary antibodies (goat anti-mouse IgG-HRP, sc-2005 and goat anti-rabbit IgG-HRP, sc-2004) for WB (Santa Cruz Biotechnology). CellTracker™ Blue (CMAC) and phalloidin (ThermoFisher). Staphylococcal enterotoxin E (SEE) (Toxin Technology, Inc).

Immunoprecipitation

Immunoprecipitation from cell lysates was performed by using Protein A/G Magnetic Beads (Pierce, ThermoScientific) following the instructions provided by the company. Briefly, 0.5 ml lysates corresponding to $2\text{--}4 \times 10^6$ transfected Jurkat cells, stimulated or not with PMA (100 ng/ml, 30 min at 37 °C), were incubated with anti-FMNL1 (clone A4, 5 μ g) for 2 h at 4° C. Subsequently, 15 μ l of magnetic beads suspension was added and incubated for 3 h at 4° C. Beads were washed 5x with lysis buffer and the antigens were eluted with 2 M glycine pH = 2 and then neutralized. Eluates were run on 6.5 % SDS-PAGE gels and proteins transferred to PVDF membranes.

Isolation and quantification of exosomes

To analyse the exosomes produced by cells transfected with the exosome reporter GFP-CD63, a similar protocol to the originally described for exosome isolation (75) was performed. We have included a 48 h post-transfection Ficoll gradient purification step to remove dead cells and cell debris, which otherwise could contaminate exosome preparations with micro and nanoparticles (2, 18). Using these standard protocols, culture supernatants from Jurkat cells transfected with GFP-CD63 or cotransfected with GFP-CD63 and with a 3-fold molar excess of the different shFMNL1-YFP plasmids, cocultured with SEE-pulsed Raji cells, were centrifuged in sequential steps to eliminate cells and cell debris/apoptotic bodies (75). Subsequently, the exosomes were recovered by ultracentrifugation (100,000xg for 12 h at 4° C) as described (16), lysed and then analysed by WB. CD63 is characteristically present in MVB, ILVs and hence in exosomes, but also in secretory lysosomes and the plasma membrane (Suppl. Video 8). Plasma membrane CD63 localization is produced by degranulation of MVB and diffusion of CD63 from the limiting membrane of MVB to the plasma membrane upon MVB fusion (2, 18) (Suppl. Video 8). Approximately $1\text{--}2 \times 10^6$ transfected Jurkat cells were challenged with SEE-pulsed Raji cells and WB signals in exosome lysates were normalized by the cell expression levels of GFP-CD63 among different transfections, cell groups and stimuli, in the WB corresponding to the cell lysates (2, 47) (18). It has been established that the exosomal CD63 WB signal in this synapse model correlates with exosome number obtained by flow cytometry (76), by electron microscopy (77) and by nanoparticle concentration analysis (nanoparticles/ml), using NTA (18). WB analysis of CD63 protein and GFP-tagged CD63 in isolated exosomes has been used as a *bona fide* method to specifically determine changes in exosome production (17, 47, 78) in

transiently-transfected Jurkat cells forming synapses with SEE-pulsed Raji cells without the detection of contaminating exosomes produced by MHC-II-stimulated Raji cells or constitutively secreted by Raji cells (79) (2, 17, 75) (18) (25) (22). No significant differences in the GFP-CD63 levels (i.e. Fig. 9B) were observed in the lysates of transfected C3 Jurkat cells, stimulated or not with SEE-pulsed Raji cells, at the end of the cell coculture period for exosome secretion, showing that the GFP-CD63⁺ exosomes were produced by an equal number of viable, GFP-CD63-expressing cells. Moreover, to quantify exosome concentration and to analyse their size distribution, the cell culture supernatant collected just before the ultracentrifugation step was diluted (1/5) in Hank's balanced salt solution (HBSS) and analysed by NTA using a NANOSIGHT equipment (LM10, Malvern) that was calibrated with 50 nm, 100 nm and 400 nm fluorescent calibration beads (Malvern). The hydrodynamic diameter measured by NTA, although apparently higher to that originally described for exosomes using electron microscopy (50–100 nm) (Fig. 9A), certainly corresponds to the real size of canonical, unfixed exosomes in solution, as described (80). The NTA measurements of exosome concentration (particles/ml) were normalized by the transfection efficiency and exosome-producing Jurkat cell number, by referring exosome concentration to GFP-CD63 signals in the WB of the cell lysates (Fig. 9B).

Western blot analysis

Cells were lysed in TritonTM X100-containing lysis buffer supplemented with both protease and phosphatase inhibitors. Approximately 50 µg of cellular proteins were recovered in the 10,000xg pellet from 10⁶ cells. Cell lysates and neutralized, acid-eluted IPs were separated by SDS-PAGE under reducing conditions and transferred to HybondTM ECLTM membranes (GE Healthcare). Membranes were incubated sequentially with the different primary antibodies and developed with the appropriate HRP-conjugated secondary antibody using enhanced chemiluminescence (ECL). When required, blots were stripped following standard protocols prior to reprobing them with primary and HRP-conjugated secondary antibodies. For exosome secretion studies, cells and isolated exosomes were lysed in RIPA lysis buffer containing protease inhibitors. Approximately 5 µg of exosomal proteins were recovered in the 100,000xg pellet from 1–2x10⁶ cells. Exosomes were resuspended in 60 µl of RIPA lysis buffer and 20 µl of exosomal or cell lysate proteins were separated by SDS-PAGE and transferred to HybondTM ECLTM membranes (GE Healthcare). For CD63 detection, proteins were separated under nonreducing conditions as described (17). For WB analysis of exosomes, each lane contained the total exosomal protein that was recovered from the culture medium coming from the same number of cells, untreated or treated with stimuli. Blots were incubated with mouse anti-CD63 (clone NKI-C-3, Oncogene) and developed with the appropriate HRP-conjugated secondary antibody using ECL. Autoradiography films were scanned and the bands were quantified using Quantity One 4.4.0 (Bio-Rad) and ImageJ (Rasband, W.S., ImageJ, National Institutes of Health, Bethesda, Maryland, USA, <http://rsb.info.nih.gov/ij/>, 1997–2004) software.

Time-lapse microscopy, immunofluorescence and image analysis

In the experiments requiring IS formation, we challenged the transfected Jurkat clones with SEE-pulsed Raji cells, a well-established synapse model (81). Raji cells were attached either to ibiTreat microwell culture dishes (ibidi) pretreated with fibronectin (0.1 mg/ml, for paraformaldehyde -PFA- fixation) or glass bottom microwell culture dishes (ibidi) pretreated with poly-L-lysine (0.02 mg/ml, for PFA and acetone fixation). Next, they were labelled with CMAC (10 µM) and pulsed with 1 µg/ml SEE and mixed with Jurkat clones transfected with the different expression plasmids, at 24–48 h posttransfection. The resulting IS were analysed as described (81) (2) (44) (26). PFA followed by acetone fixation was required for a clean FMNL1 staining and was compatible with phalloidin labelling (82). Immunofluorescence of fixed synapses was performed as previously described (83), and additional blocking and fixations steps were performed between each primary antibody and subsequent fluorochrome-coupled secondary antibody or phalloidin staining, to exclude any potential cross-reaction of secondary antibodies (i.e., Fig. 5).

For FMNL1 β relocation experiments in living cells, control (C3) and PKC δ -interfered (P5), YFP-FMNL1 β -expressing Jurkat clones were challenged with CMAC-labelled, SEE-pulsed Raji cells as described above. Wide-field, time-lapse microscopy was performed using an OKO-lab stage incubator (OKO) for sample ambient on a Nikon Eclipse TiE microscope equipped with a Prime BSI (Photometrics) digital camera, a PlanApo VC 60x/1.4NA OIL objective (Nikon) and with Perfect Focus System, allowing distinct, appropriate offsets per each XY field. Wide-field fluorescence of fixed synapses was performed by capturing 30–40 Z sections (0.3–0.4 μ m thickness) using the same microscope and NIS-AR software (Nikon), and maximum intensity projection (MIP) images of all channels were generated using NIS-AR software.

Time-lapse acquisition during the indicated times and analysis were performed using NIS-AR software (Nikon). Subsequently, in some experiments epifluorescence images were improved by Huygens Deconvolution Software from Scientific Volume Image, using the “widefield” optical option as previously described (43) (44). For quantification, digital images were analysed using NIS-AR (Nikon) or ImageJ software. The quantification and analysis of YFP MFI at the synapse area (YFP-FMNL1 β MFI IS) and at the whole cell (YFP-FMNL1 β MFI Cell) in time-lapse experiments, were performed within floating regions of interest (ROI) (i.e., ROI changing XY position over time) by using NIS-AR software (Fig. 6A and Suppl. Fig. S7).

Confocal microscopy imaging in fixed synapses was performed by using an SP8 Leica confocal microscope, with sequential acquisition, bidirectional scanning and the following laser lines: UV (405 nm, intensity: 33.4%), supercontinuum visible (633 nm, intensity: 15.2%), supercontinuum visible (550 nm, intensity: 20.8%), supercontinuum visible (488 nm, intensity: 31.2%). Deconvolution of confocal images was performed by using Huygens Deconvolution Software from Scientific Volume Image with the “confocal” optical option. 2D colocalization analyses were accomplished by using JACoP plugin from ImageJ, whereas interface colocalization and the corresponding videos to generate the IS interface were performed using the “colocalization” tool, followed by the “volume view” and then “movie maker” tools from NIS-AR. For interface colocalization analyses, the original emission colors of the different fluorochromes were changed to Red, Green or Blue, since NIS-AR requires an RGB color code for this analyses (84). In particular, F-actin and Phospho-Ser PKC substrate acquired in magenta were changed to blue, and YFP acquired in yellow was changed to green (i.e. Fig. 8, Suppl. Fig. S4, Suppl. Fig. S5 and Suppl. Fig. S10). Profile analyses of MFI corresponding to each separate channel and the colocalization pixels at the IS interface (ZX axes) were performed by using “Intensity Profile” tool in NIS-AR and then Excel, as reported (84). To measure the relative size of the F-actin low area at the cIS, which quantifies the decrease in F-actin density at the cIS, we used the last frame from the former IS interface videos (Suppl. Video 5 and Fig. 7), generated as described (84). The areas of the F-actin-low region at cIS (Fact-low cIS area) (yellow line) and the synapse (IS area) (white line) (i.e., Fig. 7) were defined in ImageJ using the ROI manager and applying an appropriate, manually-defined threshold to the F-actin channel. The definition of the ROI to measure the Fact-low cIS area and the IS area was performed manually or by using automated algorithm “auto-detect ROI/segmentation” from NIS-AR (25) (26). As for the Fact-low cIS area (yellow line) and the IS area (white line) (i.e., Fig. 7), they were defined in ImageJ using the ROI manager and applying a default threshold or, in cases where precise detection was not achieved, an appropriate, manually-defined threshold to the F-actin channel. In both cases, we have found that both manual and automated values were quite similar. The areas included in these ROIs were measured and limited to the defined thresholds, and the relative area of the F-actin-low region at the cIS (Fact-low cIS area / IS area), which is independent of both cell and synapse size, was calculated and represented (Fig. 7). Regarding the possibility that actin cytoskeleton of Raji cells can also contribute to the measurements of synaptic F-actin, it is important to remark that MHC-II-antigen triggering on the B cell side of the Th synapse does not induce noticeable F-actin changes along the synapse (i.e. F-actin clearing at the central IS), in contrast to TCR stimulation on T cell

side (85) (86) (3). In addition, we have observed that majority of F-actin at the IS belongs to the Jurkat cell (84). Thus, the contribution to the analyses of the residual, invariant F-actin from the B cell is negligible using our protocol (84).

To compare MTOC and MVB polarization in synapses, MTOC and MVB PI were calculated, as previously described (37) (25) (26) (87) (**Suppl. Fig. S2A**), using MIP, both for deconvoluted, wide-field, and confocal-acquired images. In the MIP, the positions of the cell geometric center ($Cell^C$) and MTOC and MVB center of mass ($MTOC^C$ and MVB^C , respectively) were used to project $MTOC^C$ (or MVB^C) on the vector defined by the $Cell^C$ -synapse axis. Then the MTOC (or MVB) PI was calculated by dividing the distance between the $MTOC^C$ (or MVB^C) projection and the $Cell^C$ ("B" or "A" distance, respectively) by the distance between the $Cell^C$ and the synapse ("C" distance) (**Suppl. Fig. S2A**, **Fig. 4** and **Fig. 5**). $Cell^C$ position was taken as the origin to measure distances, thus those "A" and "B" values in the opposite direction to the synapse were taken as negative. Thus, PI ($PI = A/C$ or B/C) ranked from +1 (fully polarized) to -1 (fully antipolarized). Therefore, PI values were normalized by cell size and shape (**Fig. 4**) (25) (87) and measured the relative ability of MTOC and MVB to polarize towards the IS. Remarkably, one important feature of the IS consists of both the onset of the initial cell-cell contacts and the establishment of a mature, fully productive IS, are intrinsically stochastic, rapid and asynchronous processes (88, 89) (43). Thus, the score of the PI corresponding to the distance of MTOC/MVB with respect the IS (42) may be contaminated by background MTOC/MVB polarization, in great part due to the stochastic nature of IS formation (88). In order to circumvent this caveat, a substantial number of IS of each cell group/condition were analysed to obtain statistically significant results, as reported by numerous authors (90) (37) (49) (88) (25) (26). Image analysis data correspond to at least three different experiments, analysing a minimum of 30 synapses from 15 different, randomly selected, microscopy fields per experiment. ANOVA was performed for statistical significance of the results using Excel and IBM's SPSS Statistics software.

Abbreviations:

- AICD: activation-induced cell death
- ANGPTL4: Angiopoietin-like 4
- APC: antigen-presenting cell
- BCR: B-cell receptor for antigen
- C : center of mass
- CAR T: chimeric antigen receptor-bearing T lymphocytes
- cIS: central region of the immune synapse
- CMAC: CellTracker™ Blue (7-amino-4-chloromethylcoumarin)
- cSMAC: central supramolecular activation cluster
- CTL: cytotoxic T lymphocytes
- DAD: diaphanous autoinhibitory domain
- Dia1: Diaphanous-1
- DID: diaphanous inhibitory domain
- dSMAC: distal supramolecular activation cluster
- ECL: enhanced chemiluminescence
- F-actin: filamentous actin
- Fact-low cIS: F-actin-low region at the center of the immune synapse
- FasL: Fas ligand
- FHOD1: formin homology domain protein 1
- F-actin: filamentous actin
- FMNL1: formin-like 1
- fps: frames per second

- HBSS: Hank's balanced salt solution
- HRP: horseradish peroxidase
- ILVs: intraluminal vesicles
- IS: immune synapse
- IPs: immunoprecipitates
- MFI: mean fluorescence intensity
- MHC: major histocompatibility complex
- MIP: maximum intensity projection
- MVB: multivesicular bodies
- MTOC: microtubule-organizing center
- MW: molecular weight
- NTA: Nanoparticle Tracking Analyses
- NS: not significant
- PFA: paraformaldehyde
- PKC: protein kinase C
- PKC δ : protein kinase C δ
- PMA: phorbol myristate acetate
- PI: polarization index
- pSMAC: peripheral supramolecular activation cluster
- ROI: region of interest
- ROCK: Rho-dependent protein kinase
- SD: standard deviation
- SEE: *Staphylococcal* enterotoxin E
- SMAC: supramolecular activation cluster
- SMAPs: supramolecular attack particles
- TCR: T-cell receptor for antigen
- Th: T-helper
- TRANS: transmittance
- WB: western blot
- YFP: yellow fluorescent protein.

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Supplementary information

Supplementary information is available at eLife website

Conflict of Interest

The authors report no conflict of interest.

Author Contributions Statement

V.C. and M.I. conceived and designed all the experiments. J.R-N. did most of the experiments, analysed data, and contributed to the writing of the manuscript. J.R-N., S.F. and I.S. contributed to the MTOC/MVB polarization experiments and IS image analyses. S.F. and I.S. performed the FMNL1 immunoprecipitation experiments and contributed to the FMNL1 phosphorylation studies and image analyses, and J.R-N. contributed to timelapse studies and image analyses of area F-actin. P.B. and F.R.G-G. created and sequenced the FMNL1 β mutants, and contributed to manuscript writing. M.I. conceptualized and coordinated the research, directed the study, analysed data, and wrote the manuscript. All the authors contributed to the planning and designing of the experiments and to helpful discussions.

Data Availability Statement

The materials described in the manuscript, including all relevant raw data, will be freely available to any researcher wishing to use them for non-commercial purposes, without breaching participant confidentiality.

Additional Declarations:

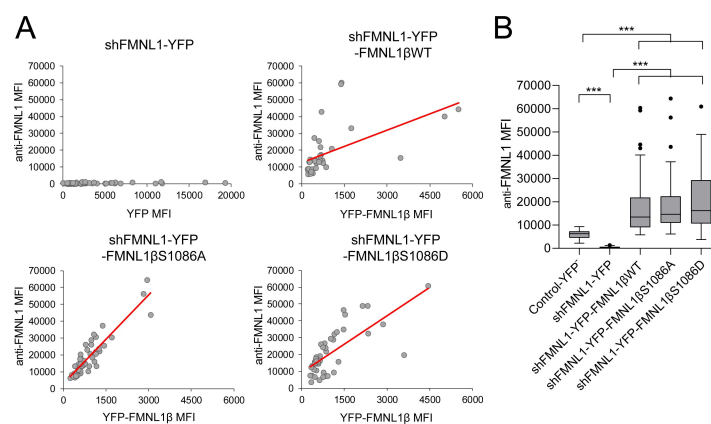
The authors declare no competing interests.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [Video1FactinMTOCenfaces.avi](#)
- [Video2FactinCD63enfaces.avi](#)
- [Video3.avi](#)
- [Video4CombinedS1086AS1086Dlabels1.avi](#)
- [Video51.avi](#)
- [Video6labelsWT.avi](#)
- [Video7labelsWT.avi](#)
- [Video8.avi](#)

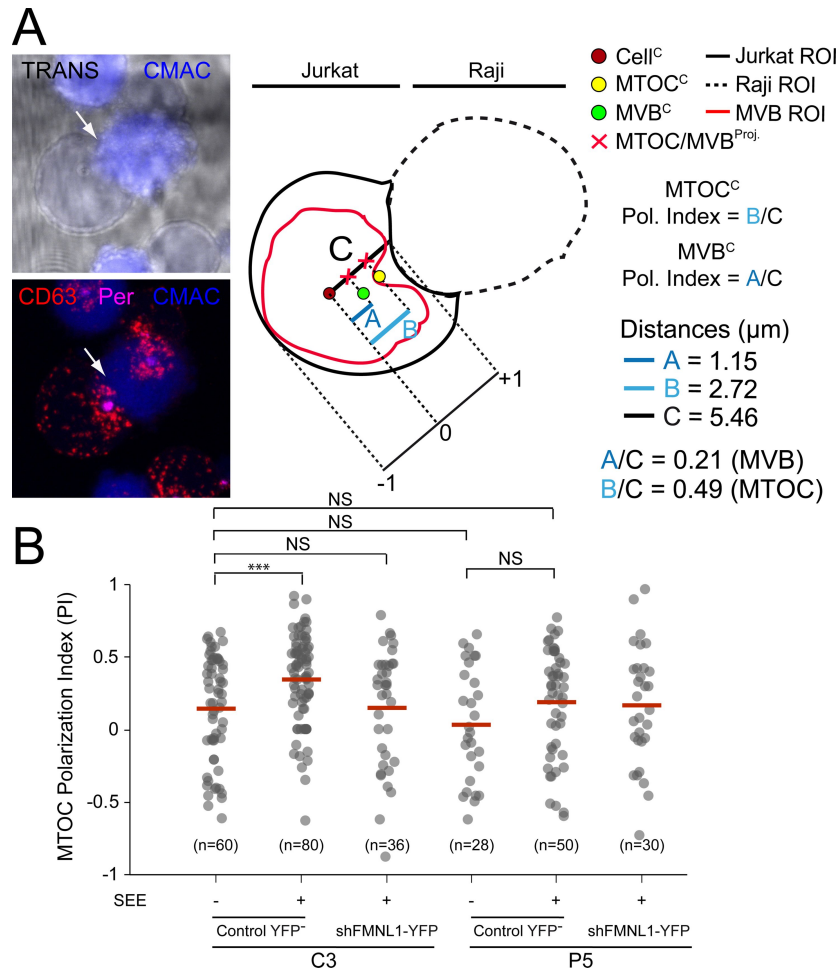
- BinderSupplFigeLife.pdf



Supplementary Fig. S1.

Expression of YFP-FMNL1 β constructs and correlation between YFP-FMNL1 β variants expression and the anti-FMNL1 antibody signal.

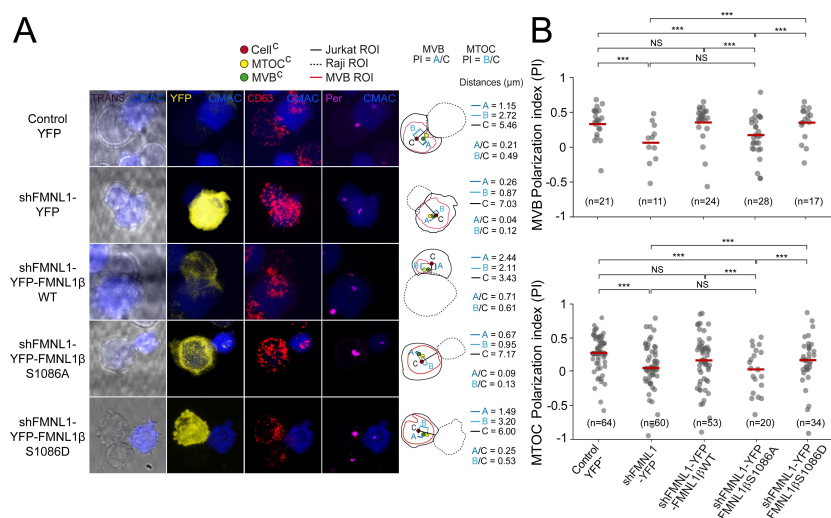
Panel A), Scatter plots of the MFI values of YFP or YFP-FMNL1 variants and the anti-FMNL1 signal in cells transfected with FMNL1-interfering (shFMNL1-HA-YFP), YFP-FMNL1 β WT (shFMNL1-HA-YFP-FMNL1 β WT), YFP-FMNL1 β S1086A, or YFP-FMNL1 β S1086D constructs are depicted. The trend line (linear regression) is displayed in red color in FMNL1 β constructions. Panel B), Box plot representing the different anti-FMNL1 MFI signals in the control and the different transfected cell groups. This figure is related to **Fig. 2**. ***, $p \leq 0.05$. Results are representative of the data from several experiments ($n=3$) with similar results.



Supplementary Fig. S2.

SEE-induced polarization of MVB and MTOC to the immune synapse and calculation of their polarization indexes.

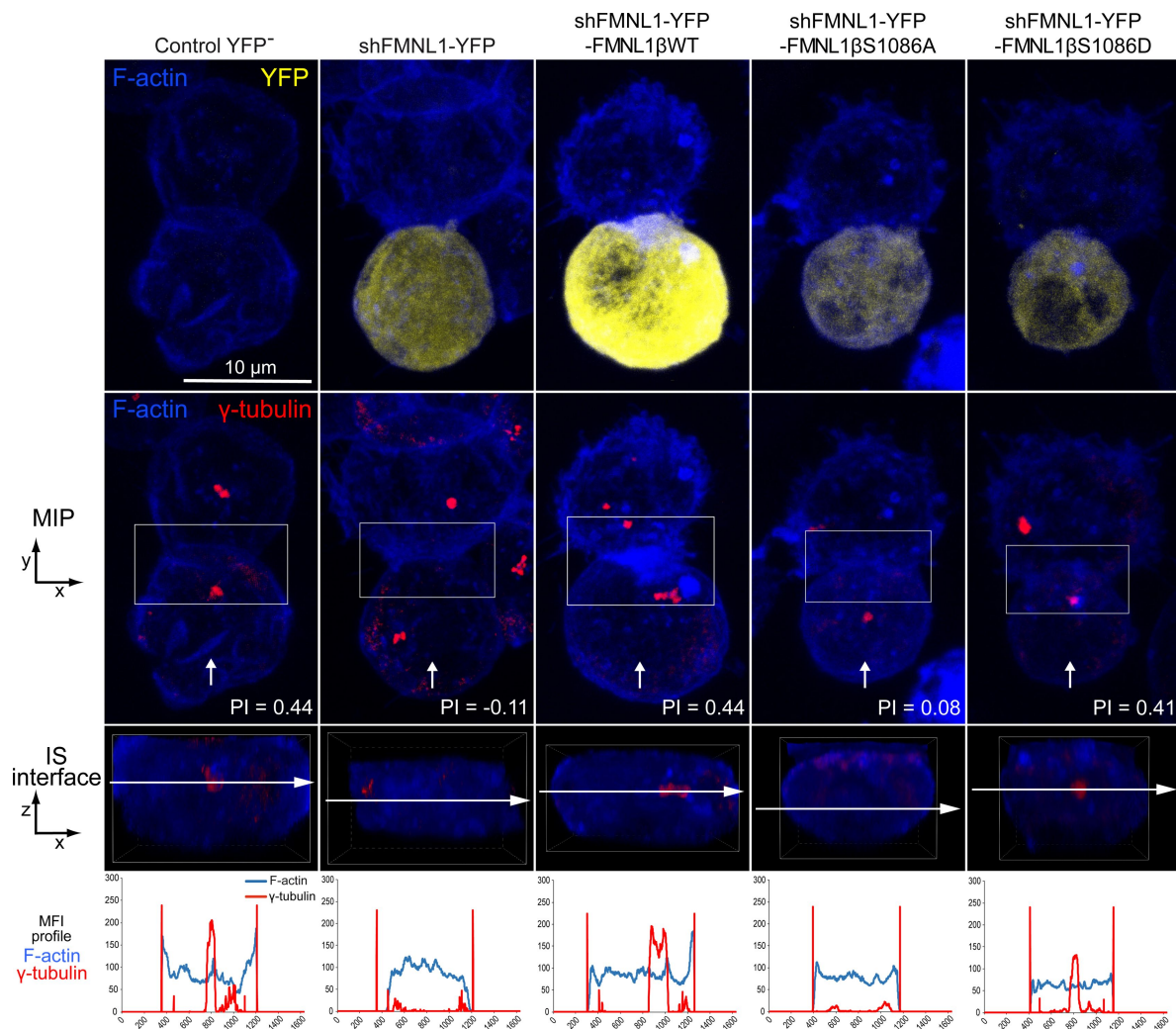
Panel A), In the left panels, synaptic cell conjugates made by Jurkat cells with CMAC-labeled, SEE-pulsed Raji cells (blue) were fixed, permeabilized and stained with anti-CD63 (red) and anti-pericentrin (magenta) antibodies to label MVB and the MTOC, respectively. The white arrow indicates the IS area. In the right-side scheme, the distances in color ("A", blue; "B", cyan and "C", black) used for the calculation of both MVB and MTOC PI (A/C and B/C, respectively) are indicated. The dark-red dot represents the cell geometric center (Cell^C), whereas the yellow and green dots indicates the MTOC and MVB center of mass (MTOC^C and MVB^C), respectively. The line "C" represents the distance between the Cell^C and the synapse, and the projections of both MTOC^C and MVB^C on the "C" line are labelled with red crosses (MTOC/MVB^{Proj}). Since the Cell^C position was taken as the origin to measure distances, those "A" or "B" values in the opposite direction to the synapse were taken as negative. Thus, PI ranged from +1 to -1. The Raji cells and the Jurkat clones are labelled with discontinuous and continuous white lines, respectively. Panel B), C3 control and PKCδ-interfered P5 clone were untransfected (Control YFP⁻) or transfected with FMNL1-interfering (shFMNL1-HA-YFP). Subsequently, cells were challenged with CMAC-labelled, unpulsed (SEE-) or SEEpulsed (SEE+) Raji cells for 1 h, fixed, stained with anti-pericentrin (magenta) to label the MTOC and imaged by epifluorescence microscopy to measure MTOC PI as indicated in panel A. The dot plot shows the MTOC PI of each indicated cell group are shown. NS, not significant; ***, $p \leq 0.05$. Results are representative of the data from several experiments ($n=3$) with similar results.



Supplementary Fig. S3.

FMNL1β phosphorylation at S1086 is involved in MTOC /MVB polarization towards the immune synapse. Confocal analysis.

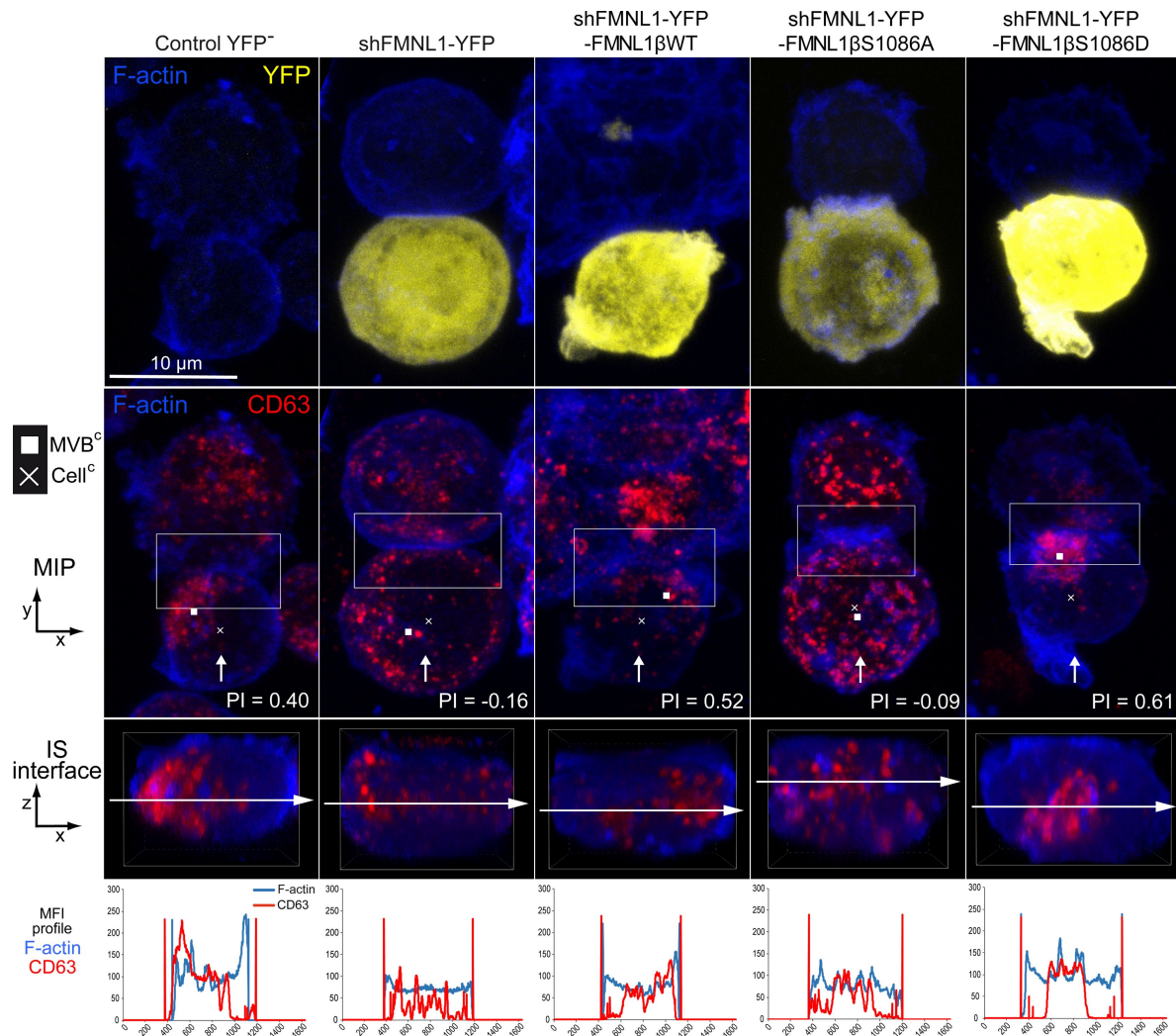
Untransfected, C3 clone (Control YFP⁺) (first row) and C3 clone transfected (yellow) with either FMNL1-interfering (shFMNL1-HA-YFP) (second row) or FMNL1-interfering and expressing interference-resistant YFP-FMNL1βWT (shFMNL1-HA-YFP-FMNL1βWT) (third row), or YFP-FMNL1βS1086A (fourth row), or YFP-FMNL1βS1086D (fifth row) constructs, were challenged with CMAC-labelled, SEE-pulsed Raji cells (blue) for 1 h to form IS. Then, cell conjugates were fixed, stained with anti-pericentrin to label the MTOC (magenta) and anti-CD63 to label MVB (red), and imaged by confocal fluorescence microscopy. Panel A), representative MIP images with merged channels of each cell group specified in the left side, and the diagrams on the right represent the measurements of the parameters used to calculate the MTOC and MVB PI, including the distance between the IS and the center of the cell (Cell^C) and the IS (line C), the MTOC^C (line B) and the MVB^C (line A). Jurkat cell outlines are labelled with a continuous line, Raji cells are labelled with a dashed line, whereas ROI containing MVB are labelled with a continuous red line. Panel B), MTOC and MVB PI dot plot for the different cell groups corresponding to the indicated number of synapses from a similar experiment to that described in panel A) are represented. This figure is related to **Fig. 4**. NS, not significant. ***, $p \leq 0.05$. Results are representative of the data from several experiments (n=3) with similar results.



Supplementary Fig. S4.

T cell *en face* analysis of F-actin and MTOC distribution at the immune synapse interface.

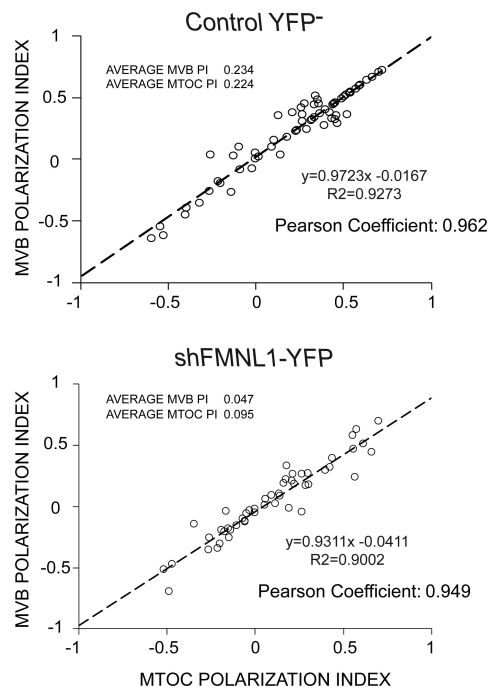
C3 control clone was untransfected (Control YFP⁻) (first column) or transfected with FMNL1-interfering (shFMNL1-HA-YFP) (second column), or FMNL1-interfering expressing interference-resistant YFP-FMNL1 β WT (shFMNL1-HA-YFP-FMNL1 β WT) (third column), YFP-FMNL1 β S1086A (fourth column), or YFP-FMNL1 β S1086D (fifth column) constructs. Subsequently, cells were challenged with CMAC-labelled SEEpulsed Raji cells (blue) for 1 h, fixed, stained with phalloidin (magenta) and anti-Y-tubulin (red). The corresponding shFMNL1 constructions are in yellow, and cells were imaged by confocal fluorescence microscopy. Please realize that F-actin (acquired in magenta) was changed to blue color. The upper row includes the top, yx views corresponding to the MIP images of representative examples of each transfected cell group. In the second row, the same examples are displayed, showing the relative MTOC position of each transfected cell group and MTOC PI is indicated in white types. Vertical white arrows indicate the direction to visualize the *en face* views of the IS (IS interface) enclosed by the ROIs (white rectangles), as shown in [Suppl. Video 1](#). Subsequently, in the third row the *en face* zx views were generated by merging the indicated channels in the second row of each panel (F-actin in blue merged to anti-Y-tubulin in red), on the IS interfaces of the synaptic areas (generated as shown in [Suppl. Video 1](#)). The last frame of this video corresponds to *en face* view (interface) (third row). MFI profiles along the indicated line (horizontal white arrow) of each separate channel (F-actin in blue, anti-Y-tubulin in red) are shown below the IS interfaces. This figure is related to [Fig. 4](#) and [7](#). Results are representative of data from several experiments (n=3) with similar results.



Supplementary Fig. S5.

T cell *en face* analysis of F-actin and MVB distribution at the immune synapse interface.

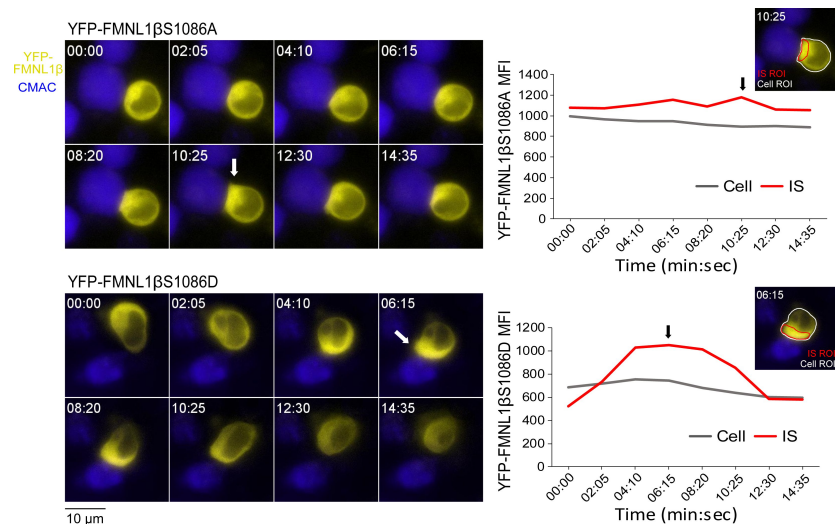
C3 control clone was untransfected (Control YFP⁻) (first column) or transfected with FMNL1-interfering (shFMNL1-HA-YFP) (second column), or FMNL1-interfering expressing interference-resistant YFP-FMNL1βWT (shFMNL1-HA-YFP-FMNL1βWT) (third column), YFP-FMNL1βS1086A (fourth column), or YFP-FMNL1βS1086D (fifth column) constructs. Subsequently, cells were challenged with CMAC-labelled SEE-pulsed Raji cells (blue) for 1 h, fixed, stained with phalloidin (magenta) and anti-CD63 (red). The corresponding shFMNL1 constructions are in yellow, and cell conjugates were imaged by confocal fluorescence microscopy. Please realize that F-actin (acquired in magenta) was changed to blue color. The upper row includes the top, yx views corresponding to the MIP images of representative examples of each transfected cell group. In the second row, the same examples are displayed, showing the MVB position in each transfected cell group and MVB PI is indicated in white types. Vertical white arrows indicate the direction to visualize the *en face* views of the IS (IS interface) enclosed by the ROIs (white rectangles), as shown in **Suppl. Video 2**. The white cross represents the position of the geometric center of the Jurkat cell (Cell^C), while the white square represents the position of the center of mass of the distribution of MVBs (MVB^C). Subsequently, in the third row, the *en face* zx views were generated by merging the indicated channels in the second row of each panel (F-actin in blue merged to anti-CD63 in red), on the IS interfaces of the synaptic areas (generated as shown in **Suppl. Video 2**). The last frame of these videos corresponds to *en face* views (interface) (third row). MFI profiles along the indicated line (horizontal white arrow) of each separate channel (F-actin in blue, anti-CD63 in red) are shown below the IS interfaces. When MVB are not polarized, some MVB can still be observed at cIS because they are scattered throughout the cell. This figure is related to **Fig. 4** and **7**. Results are representative of the data from several experiments (n=3) with similar results.



Supplementary Fig. S6.

Correlation between MTOC and MVB polarization indexes.

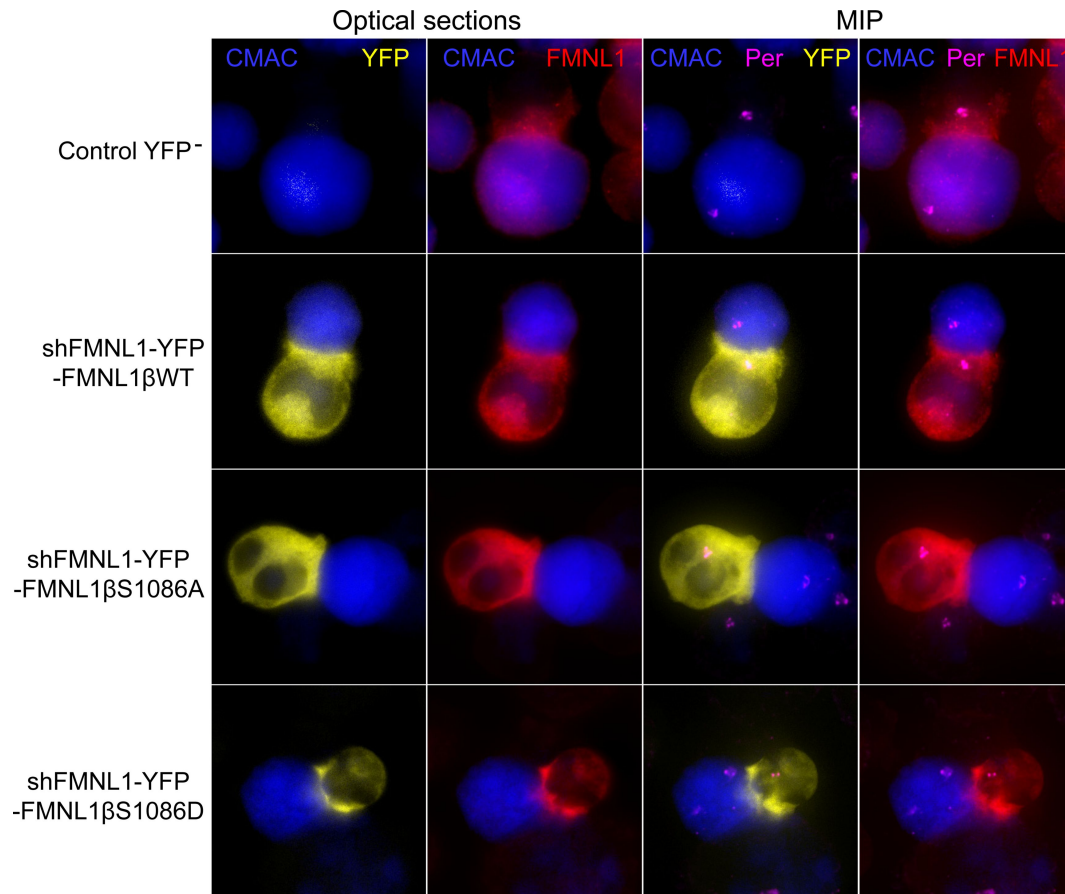
C3 control untransfected (Control YFP⁻) and C3 transfected with the FMNL1-interfering plasmid (shFMNL1-HA-YFP) were challenged with CMAC-labelled SEE-pulsed Raji cells for 1 h, fixed, stained with anti- γ -tubulin AF647 to label MTOC and anti-CD63 AF546, to label MVB and imaged by confocal fluorescence microscopy. Subsequently, MTOC and MVB PI were calculated for the synapses established by both cell groups, as described in Materials and Methods and shown in schematic representation in **Suppl. Fig. S2A**. Linear correlation analyses between the MTOC and MVB PI for both cell groups are represented, as well as the corresponding linear correlation analysis data and the Pearson's coefficients. Dashed lines represent the adjusted, regression line. This figure is related to **Fig. 4**. Results are representative of data from several experiments (n=3) with similar results.



Supplementary Fig. S7.

YFP-FMNL1βS1086A and S1086D mutants are recruited to the immune synapse.

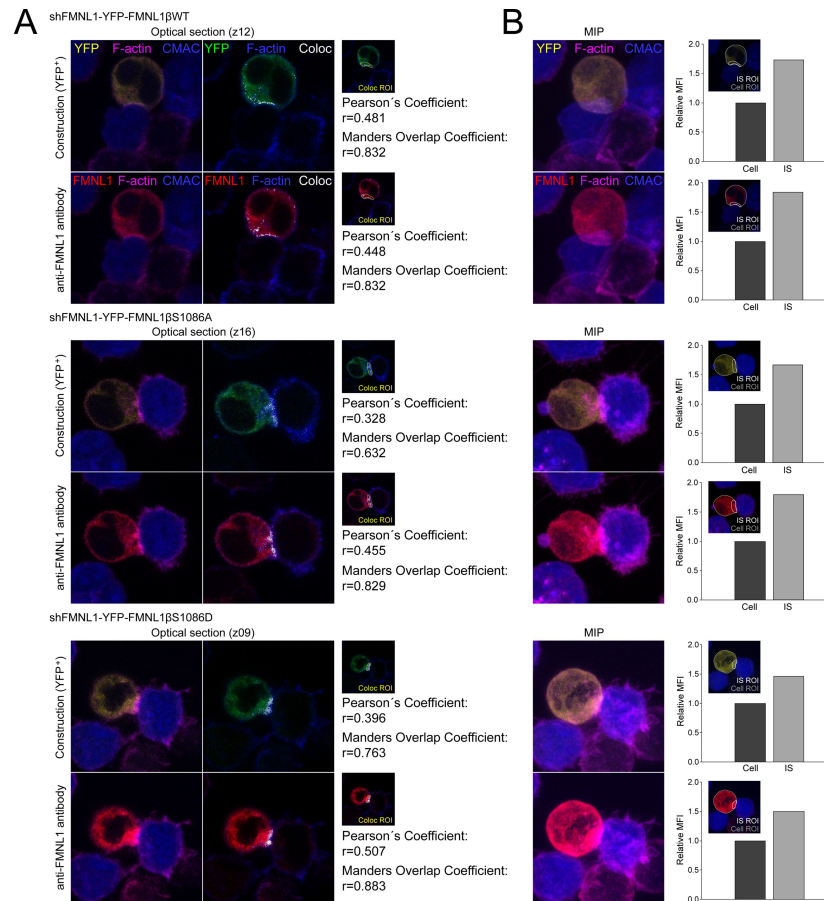
C3 control cells were transfected with FMNL1-interfering expressing interferenceresistant YFP-FMNL1βS1086A or S1086D (shFMNL1-HA-YFP-FMNL1βS1086A or shFMNL1-HA-YFP-FMNL1βS1086D) plasmids. Subsequently, transfected cells were simultaneously challenged with CMAC-labelled, SEE-pulsed Raji cells (blue) attached to slides and time-lapse acquisition of emerging synapses was performed as indicated in Materials and Methods. The videos (7 fps) (**Suppl. Video 4**) were captured and in the left side, representative frames from each video are shown. White arrows indicate accumulations of YFP-FMNL1βS1086A or S1086D at the IS. In the right side, YFP-FMNL1βS1086A or S1086D MFI within the cell ROI (grey line) and the IS ROI (red line) are represented. The inserts in the diagrams include the cell ROI (white) and the IS ROI (red) used for the time-lapse measurements on representative frames for both FMNL1β mutants. This figure is related to **Fig. 6** and **Suppl. Video 4**. At least 6 synapses of each cell group were analysed. Results are representative of data from several experiments (n=3) with similar results.



Supplementary Fig. S8.

FMNL1 and FMNL1 β accumulation at the immune synapse revealed by epifluorescence microscopy.

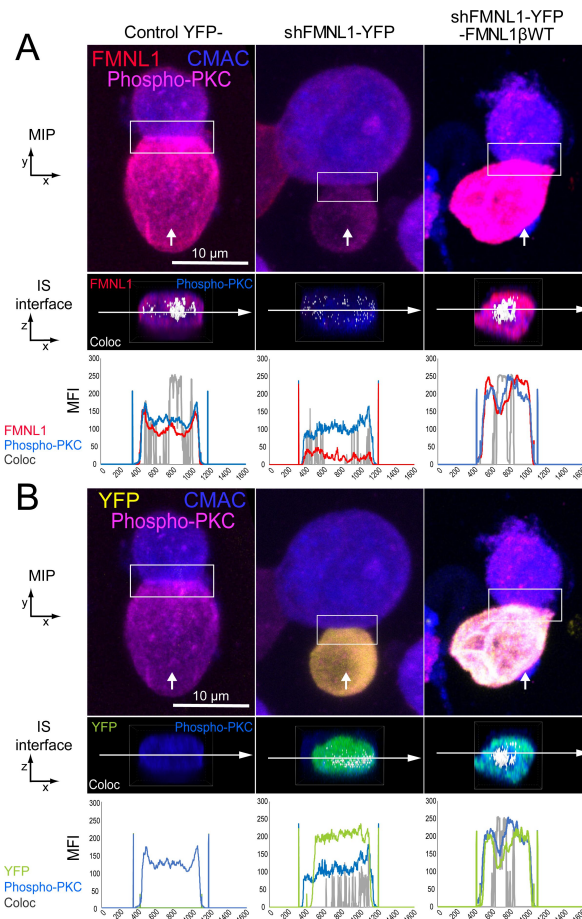
C3 control clone was untransfected (Control YFP⁻) (first row) or transfected with FMNL1-interfering expressing interference-resistant YFP-FMNL1 β WT (shFMNL1- HA-YFP-FMNL1 β WT) (second row) or YFP-FMNL1 β S1086A (third row) or YFP-FMNL1 β S1086D (fourth row) constructs. Subsequently, cells were challenged with CMAC-labelled SEE-pulsed Raji cells (blue) for 1 h, fixed, stained with anti-pericentrin (magenta) and anti-FMNL1 (red) and imaged by epifluorescence microscopy. The corresponding FMNL1 β constructions are in yellow, whose signal is observed in both the first and third columns. The first and second columns represent those individual optical sections in which the accumulation of FMNL1 or FMNL1 β variant in the IS formed by each cell group is better visualized, while the third and fourth columns represent MIPs of the same selected cells. FMNL1 or FMNL1 β variant accumulation at the IS can be visualized in each cell group in red or yellow, respectively. This figure is related to **Fig. 6**. Results are representative of the data from several experiments (n=3) with similar results.



Supplementary Fig. S9.

YFP-FMNL1 β variants accumulation at the immune synapse revealed by confocal fluorescence microscopy.

C3 control clone was transfected with FMNL1-interfering expressing interferenceresistant YFP-FMNL1 β WT (shFMNL1-HA-YFP-FMNL1 β WT) (first and second rows), YFP-FMNL1 β S1086A (third and fourth rows) or YFP-FMNL1 β S1086D (fifth and sixth rows) constructs. Subsequently, cells were challenged with CMAC-labelled SEE-pulsed Raji cells (blue) for 1 h, fixed, stained with phalloidin (magenta) and anti-FMNL1 (red). The corresponding shFMNL1 constructions are in yellow, and cells were imaged by confocal fluorescence microscopy. Panel A), top views corresponding to representative examples of each transfected cell group are shown. The first column includes those individual optical sections in which the accumulation of each FMNL1 β variant in the IS formed by each cell group is better visualized. Please realize that, since colocalization in NIS-AR only works with red, green and blue channels, F-actin (acquired in magenta) was changed to blue color, and YFP (acquired in yellow) was changed to green in the second column of each panel, where colocalization pixels between anti-FMNL1 or YFP-FMNL1 β and F-actin in the same optical section are shown in white. The third column displays the ROIs used to measure the indicated colocalization coefficients. The Pearson and Manders colocalization coefficients for each of the selected cells are displayed. Panel B), the MIPs of the same cells displayed in panel A) are shown. At the right side, the relative MFI quantification of the accumulation at the IS of YFP-FMNL1 β variants and FMNL1 compared to the signal of the entire cell is shown. This figure is related to [Fig. 6](#). Results are representative of data from several experiments (n=3) with similar results.



Supplementary Fig. S10.

Interface colocalization of FMNL1 and anti-phospho-Ser PKC substrate at the immune synapse interface.

Untransfected, control C3 clone (Control YFP⁻) (first column) and C3 clone transfected with either FMNL1-interfering (shFMNL1-HA-YFP) (second column), or FMNL1-interfering and expressing interference-resistant YFP-FMNL1βWT (shFMNL1-HA-YFP-FMNL1βWT) (third column) constructs were challenged with CMAC-labelled SEE-pulsed Raji cells (blue) for 1 h, fixed, stained with anti-Phospho-Ser PKC substrate (magenta) and anti-FMNL1 (red), and imaged by confocal fluorescence microscopy. Please realize that interface colocalization in NIS-AR only works with red, green and blue channels, therefore anti-Phospho-Ser PKC substrate (acquired in magenta) was changed to blue color, and YFP (acquired in yellow) was changed to green, for this purpose. Panel A), The upper row includes representative examples of the top yx views corresponding to the MIP images of the indicated, merged channels of each cell group. The white arrows indicate the direction to visualize the *en face* zx views of the IS (IS interface) enclosed by the ROIs (white rectangles), as shown in [Suppl. Video 7](#). In the second row, the enlarged ROIs (2x zoom) used to generate the IS interface images of each cell group are shown. Subsequently, the colocalization pixels (white) along all the Z stacks were generated by merging the indicated channels in the second row of each panel (anti-phospho-Ser PKC substrate in blue merged to anti-FMNL1 in red), on the IS interfaces of the synaptic areas (generated as shown in [Suppl. Video 7](#)). The last frame of this video corresponded to *en face* view (interface) (second row on each panel). MFI profile along the indicated line (horizontal white arrow) of each separate channel (anti-phospho-Ser PKC substrate in blue and anti-FMNL1 in red) and the colocalization pixels (grey) are shown below the IS interfaces. Panel B), same as panel A), but both the top views and the IS interfaces were generated and stained with anti-phospho-Ser PKC substrate (blue) merged to YFP (green). At least 8 synapses of each cell group were analysed. Results are representative of data from several experiments (n=3) with similar results.

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Editors

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Reviewer #1 (Public Review):

Summary:

In their article entitled "Formin-like 1 beta phosphorylation at S1086 is necessary for secretory polarized traffic of exosomes at the immune synapse", Javier Ruiz-Navarro and co-workers address the question of the mechanisms regulating the polarization of the microtubule organizing center (MTOC) and of the multivesicular bodies (MVB) at the immunological synapse (IS) in T lymphocytes.

This work is a follow-up of previous studies published by the same team showing that TCR-stimulated protein kinase C delta(PKCdelta) phosphorylates FMNL1beta, which plays a crucial role in cortical actin reorganization at the IS, and controls MTOC/MVB polarization and thus exosome secretion by T lymphocytes at the IS.

The authors first compare the amino acid sequences of FMNL2 and of FMNL1beta, to seek similarities in the DID-DAD auto-inhibition sequences and find that the sequence surrounding S1086 in the arginine-rich DAD of FMNL1beta displays high similarity to that around S1072 in FMNL2 which is phosphorylated by PKCdelta. They then interrogate the role of the phosphorylation of S1086 in the arginine-rich DAD of FMNL1beta by introducing S1086A and S1086D mutations that, respectively, cannot be phosphorylated or mimic the phosphorylated form of FMNL1beta, in cells expressing an FMNL1 shRNA.

Using these tools, they show that:

- FMNL1beta is phosphorylated by PMA an activator of PKCs.
- The S1086A mutant of FMNL1beta does not restore the defect in MTOC and MVB polarization at the IS present in FMNL1 deficient T cells, whereas the phosphomimetic mutant does.
- Although FMNL1beta phosphorylation at S1086 is necessary, it is not sufficient for MTOC polarization, since it does not restore the defect of polarization observed in PKCdelta deficient T cells.
- FMNL1b translocates to the IS. This neither requires PKC expression nor phosphorylation of S1086.
- Phosphorylation of FMNL1beta on S1086 regulates actin remodeling at the immune synapse.
- Phosphorylation of FMNL1beta on S1086 regulates secretion of extracellular vesicles containing CD63 by T lymphocytes.

Strengths:

This work shows for the first time the role of the phosphorylation of FMNL1beta on S1086 on the regulation of the IS formation and secretion of extracellular vesicles by T lymphocytes.

Weaknesses:

Although of interest, this work has several weaknesses. First, all the experiments are performed in Jurkat T cells that may not recapitulate the regulation of polarization in primary T cells. Moreover, all the experiments analyzing the role of PKCdelta are performed in one clone of wt or PKCdelta KO Jurkat cells. This is problematic since clonal variation has been reported in Jurkat T cells. Moreover, the remodeling of F-actin at the IS lacks careful quantification as well as detailed analysis of the actin structure in mutant cells. Finally, although convincing, the defect in the secretion of vesicles by T cells lacking phosphorylation of FMNL1beta on S1086 is preliminary. It would be interesting to analyze more precisely this defect. The expression of the CD63-GFP in mutants by WB is not completely convincing. Are other markers of extracellular vesicles affected, e.g. CD3 positive?

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Reviewer #2 (Public Review):

Summary:

The authors have addressed the role of S1086 in the FMNL1beta DAD domain in F-actin dynamics, MVB polarization, and exosome secretion, and investigated the potential implication of PKCdelta, which they had previously shown to regulate these processes, in FMNL1beta S1086 phosphorylation. This is based on:

- (1) the documented role of FMNL1 proteins in IS formation;
- (2) their ability to regulate F-actin dynamics;
- (3) the implication of PKCdelta in MVB polarization to the IS and FMNL1beta phosphorylation;
- (4) the homology of the C-terminal DAD domain of FMNL1beta with FMNL2, where a phosphorylatable serine residue regulating its auto-inhibitory function had been previously identified.

They demonstrate that FMNL1beta is indeed phosphorylated on S1086 in a PKCdelta-dependent manner and that S1086-phosphorylated FMNL1beta acts downstream of PKCdelta to regulate centrosome and MVB polarization to the IS and exosome release. They provide evidence that FMNL1beta accumulates at the IS where it promotes F-actin clearance from the IS center, thus allowing for MVB secretion.

Strengths

The work is based on a solid rationale, which includes previous findings by the authors establishing a link between PKCdelta, FMNL1beta phosphorylation, synaptic F-actin clearance, and MVB polarization to the IS. The authors have thoroughly addressed the working hypotheses using robust tools. Among these, of particular value is an expression vector that allows for simultaneous RNAi-based knockdown of the endogenous protein of interest (here all FMNL1 isoforms) and expression of wild-type or mutated versions of the protein as YFP-tagged proteins to facilitate imaging studies. The imaging analyses, which are the core of the manuscript, have been complemented by immunoblot and immunoprecipitation studies, as well as by the measurement of exosome release (using a transfected MVB/exosome reporter to discriminate exosomes secreted by T cells).

Weaknesses

The data on F-actin clearance in Jurkat T cells knocked down for FMNL1 and expressing wild-type FMNL1 or the non-phosphorylatable or phosphomimetic mutants thereof would need to be further strengthened, as this is a key message of the manuscript. Also, the entire work has

been carried out on Jurkat cells. Although this is an excellent model easily amenable to genetic manipulation and biochemical studies, the key finding should be validated on primary T cells.

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Author response:

We are planning to extend our results of the Jurkat model system to primary T cells, as requested by the referees and eLife's Senior Editor. This will involve the inclusion of new figures, including super-resolution/STED images to reinforce our results and to satisfy the referees' points. In addition, we will improve and/or replace all the mentioned images to solve the raised caveats, including further quantification and analyses.